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MOHSIN IBNA HOSSAIN
AIUB, MATH-6 Notes.

COMPUTATIONAL STATISTICS & PROBABILITY

#DATA VISUALIZATION

Ch:-01

#Data Visualization:-

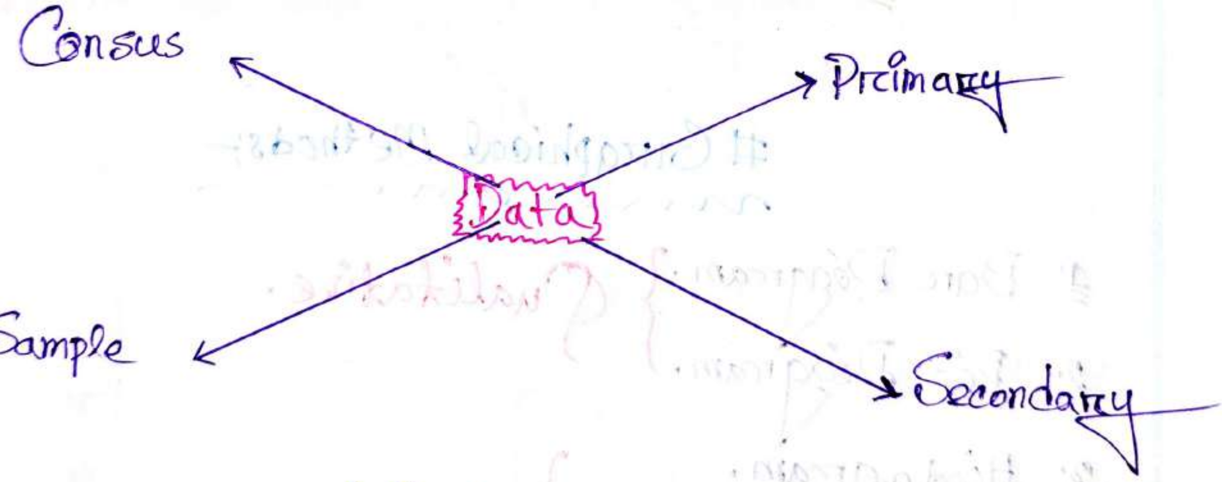
Population:-

⇒ It's consists of all items or units which are under investigation in a statistical study. denoted by N

Sample:-

⇒ It's a representative part of the population units from which information are to be collected. denoted by n

Population(N) $\geq$ Sample(n)



#Variable:-

⇒ The characteristic which varies from one unit to another is called a variable.

⇒ There are two types:-

1. Qualitative → e.g: Color, gender, name etc
2. Quantitative → e.g: weight, time, height, speed etc.

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⇒ Quantitative Variable are further classified as:-

1. Discrete Variable.
2. Continuous Variable.

Data Representation:-

⇒ There are two types.

1. Graphical/Diagrams methods.
2. Tabulation method.

Graphical Methods:-

1. Bar Diagram.
2. Pie Diagram.

} Qualitative.

3. Histogram.
4. Frequency Curve.
5. Scatter Diagram.

} Quantitative.

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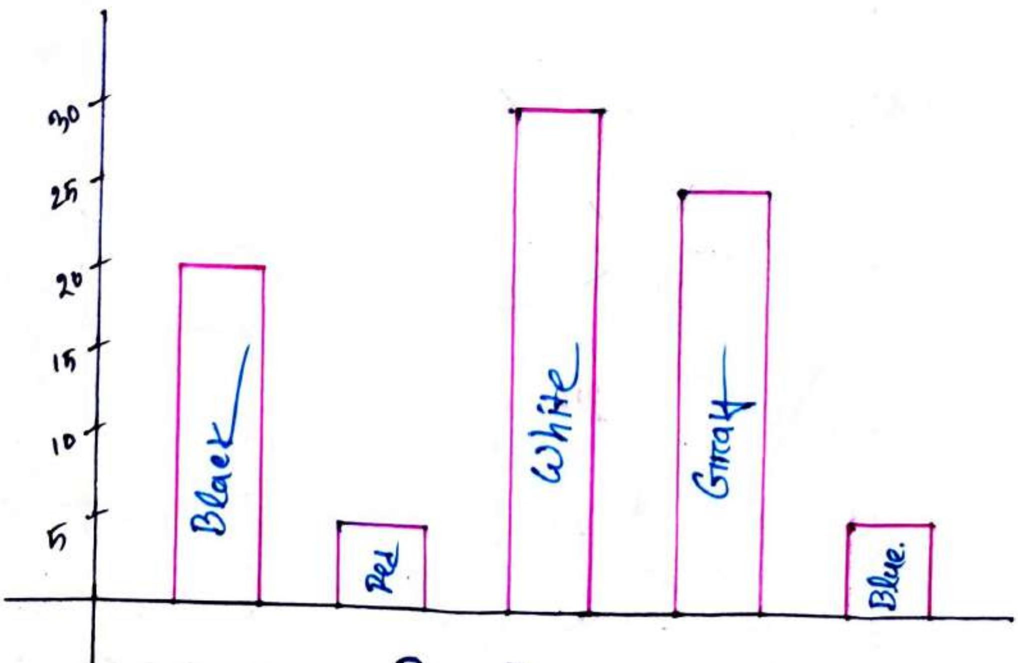
Example:- # Bar & Pie Diagram:-

Colors of Cars	No. of cars.
Black	20
Red	05
White	30
Gray	25
Blue	06
total	86

(a) Draw the bar diagram of the data.

(b) Represent the data by a pie diagram.

(a)



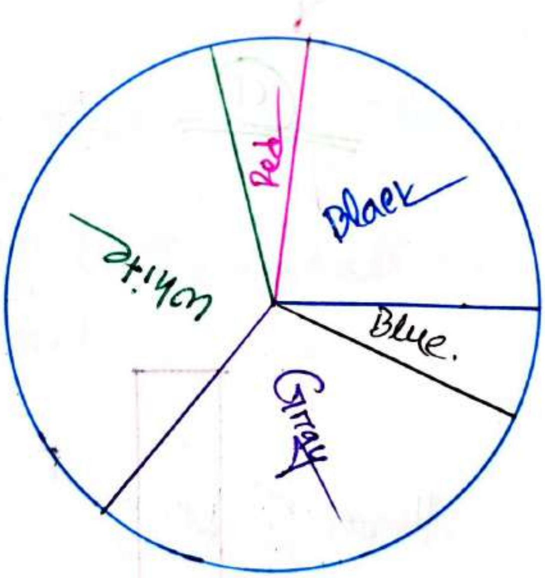
Bar Diagram.

GOOD LUCK

(b)

⇒ For a Pie diagram, we need to calculate the angle.

Colors of cars	No. of Cars.	Angles = $\frac{x \times 360}{total}$
Black	20	$= \frac{20 \times 360}{86} = 83.72 \approx 84$
Red	05	$= \frac{05 \times 360}{86} = 20.93 \approx 21$
White	30	$= \frac{30 \times 360}{86} = 125.58 \approx 126$
Gray	25	$= \frac{\cancel{104.65}\,25 \times 360}{86} = 104.65 \approx 104$
Blue	06	$= \frac{06 \times 360}{86} = 25.11 \approx 25$
total	86	



pie Diagram.

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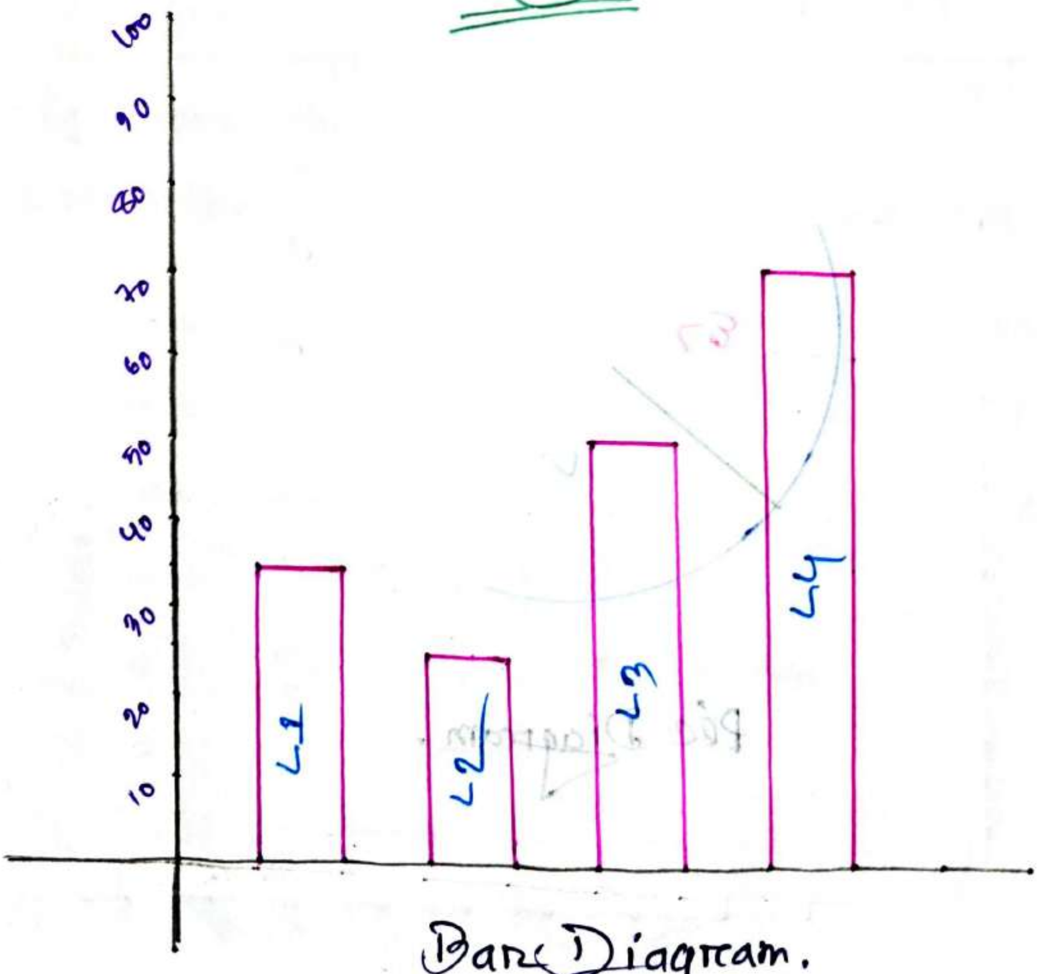
Example: 1.8

Laboratory	No. of Computer
L1	35
L2	25
L3	50
L4	70
Total	180

(a) Draw a bar diagram.

(b) Draw a pie diagram.

(a)



Bar Diagram.

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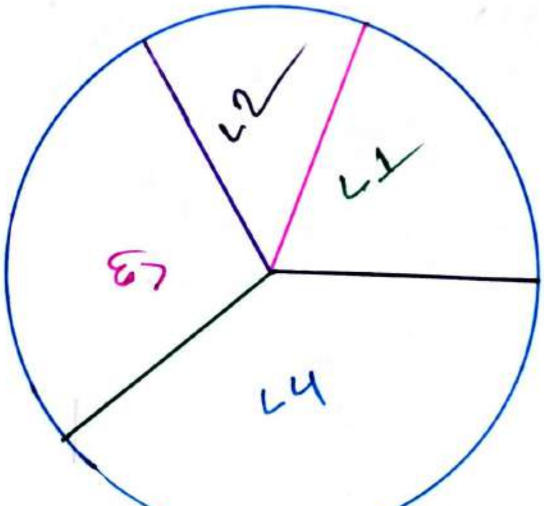
(b)

⇒ Now, we need to calculate the angle.

Name of Labs	No. of Comp.	Angles $= \frac{x \times 360}{total}$
L1	35	70
L2	25	50
L3	50	100
L4	70	140
total	180	



Pée Diagram

"GOOD LUCK"

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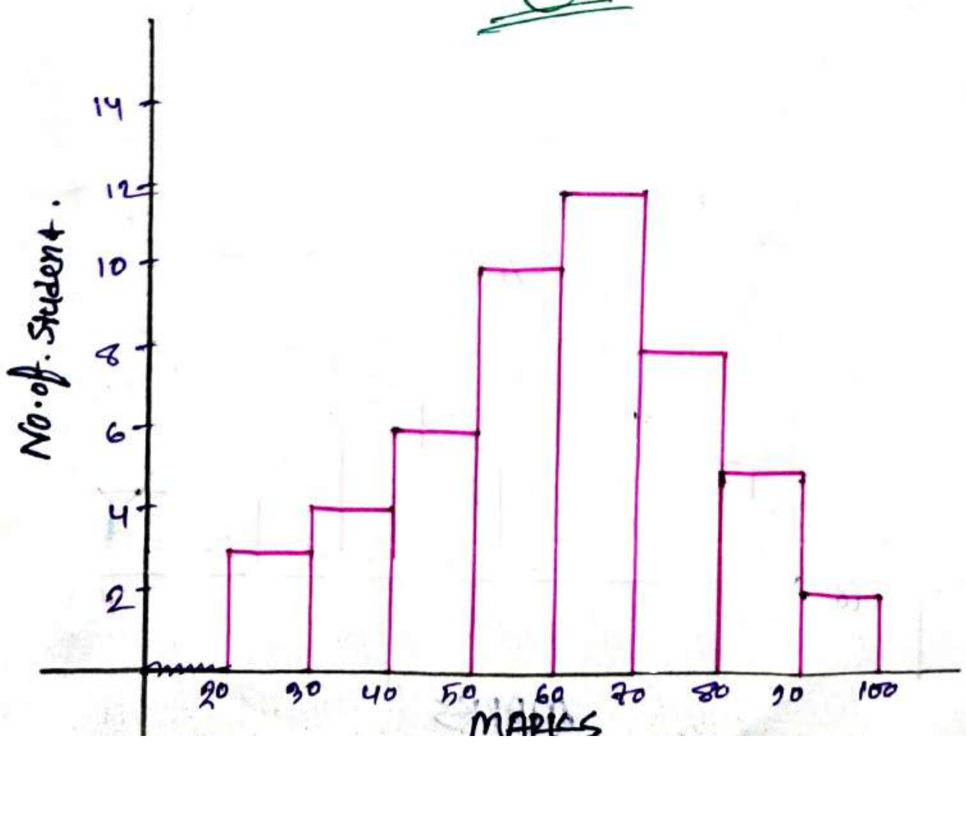
Histogram & Frequency Curve:-

#Exe 1.1:-

Marks	No. of. Student
20-30	3
30-40	4
40-50	6
50-60	10
60-70	12
70-80	8
80-90	5
90-100	2
total	40

(a) Represent the data by histogram
(b) Draw the frequency curve of the data

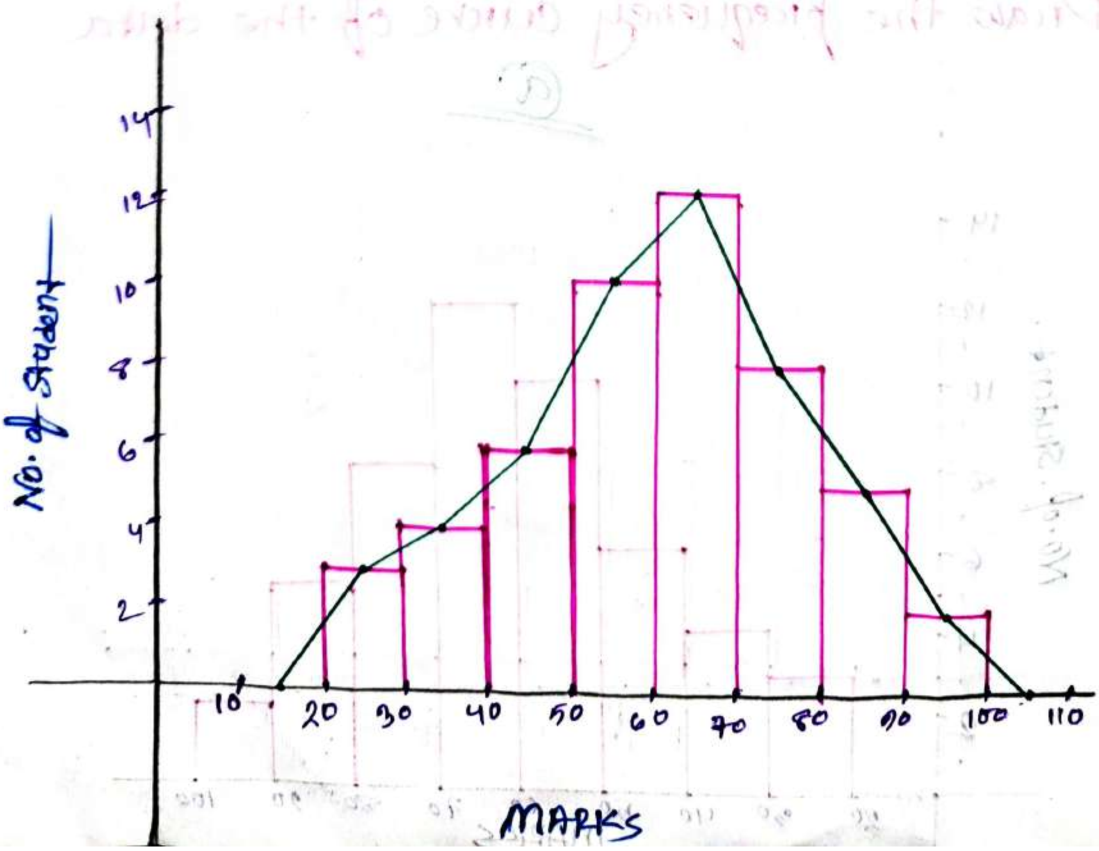
(a)



(b)

⇒ For Frequency Curve, we need to calculate the mid point of marks.

Marks	MID value	No. of Student
20-30	25	3
30-40	35	4
40-50	45	6
50-60	55	10
60-70	65	12
70-80	75	8
80-90	85	5
90-100	95	2
total		40



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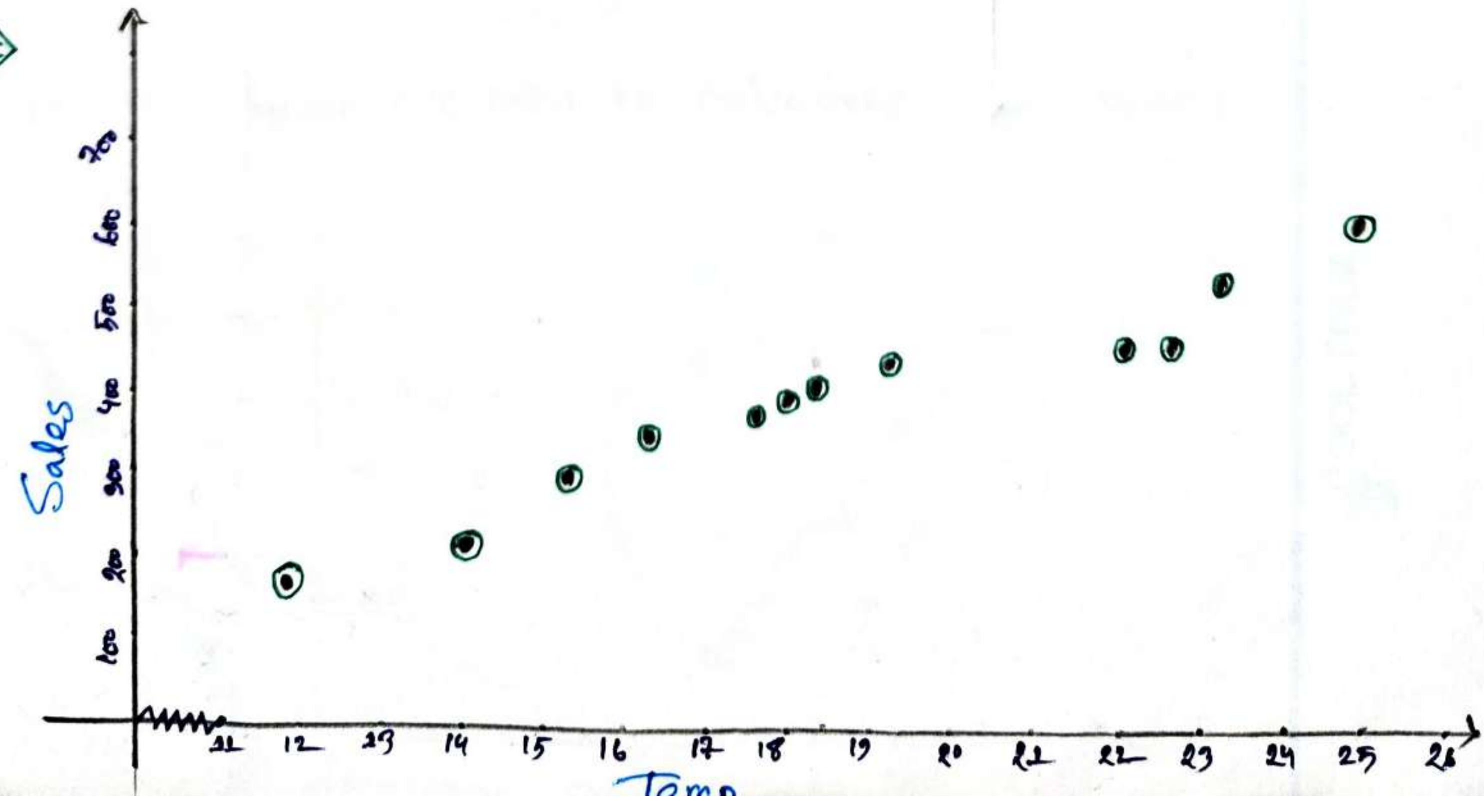
#Scatter Diagrams—

#Example 1.12 :-

Temperature °C (x)	Ice Cream Sales (y)
14.2°	215
16.4°	332
11.9°	185
15.2°	325
18.5°	421
22.1°	445
19.4°	432
25.1°	614
23.4°	524
18.1°	408
22.6°	445
17.2°	406

⇒



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Exercise:—

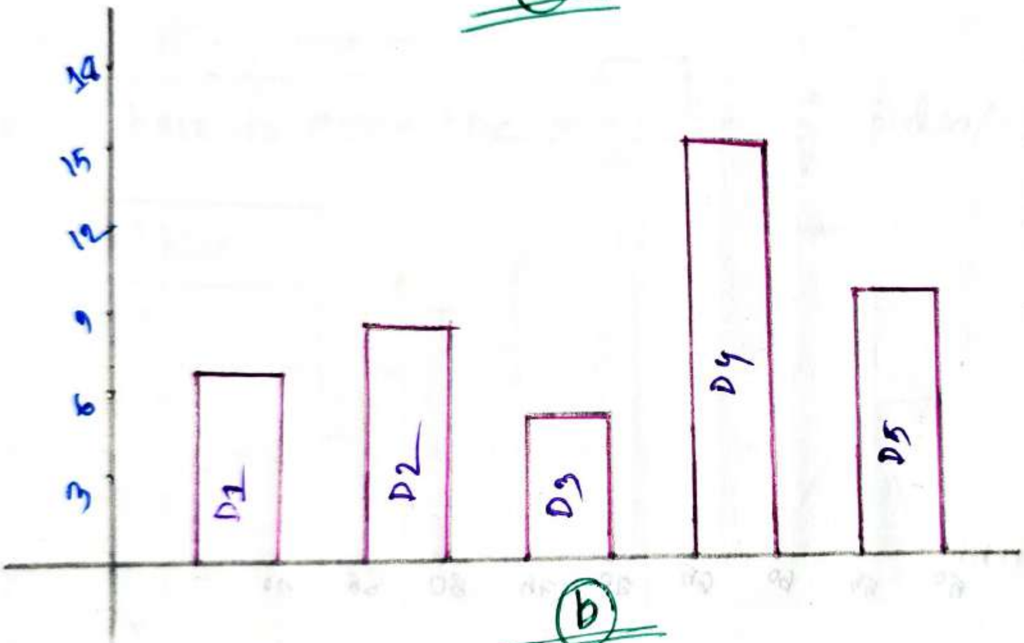
Q:2.4:—

Days	1	2	3	4	5	Total
No. of Calls	7	8	5	15	10	45

(a) Draw bar diagram.

(b) Draw pie diagram.

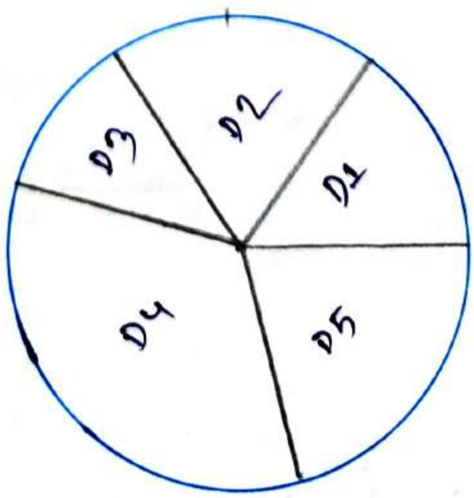
(a)



(b)

⇒ For pie diagram we need to calculate the angels.

$Angle = \frac{x \times 360}{total}$
56
64
40
120
80

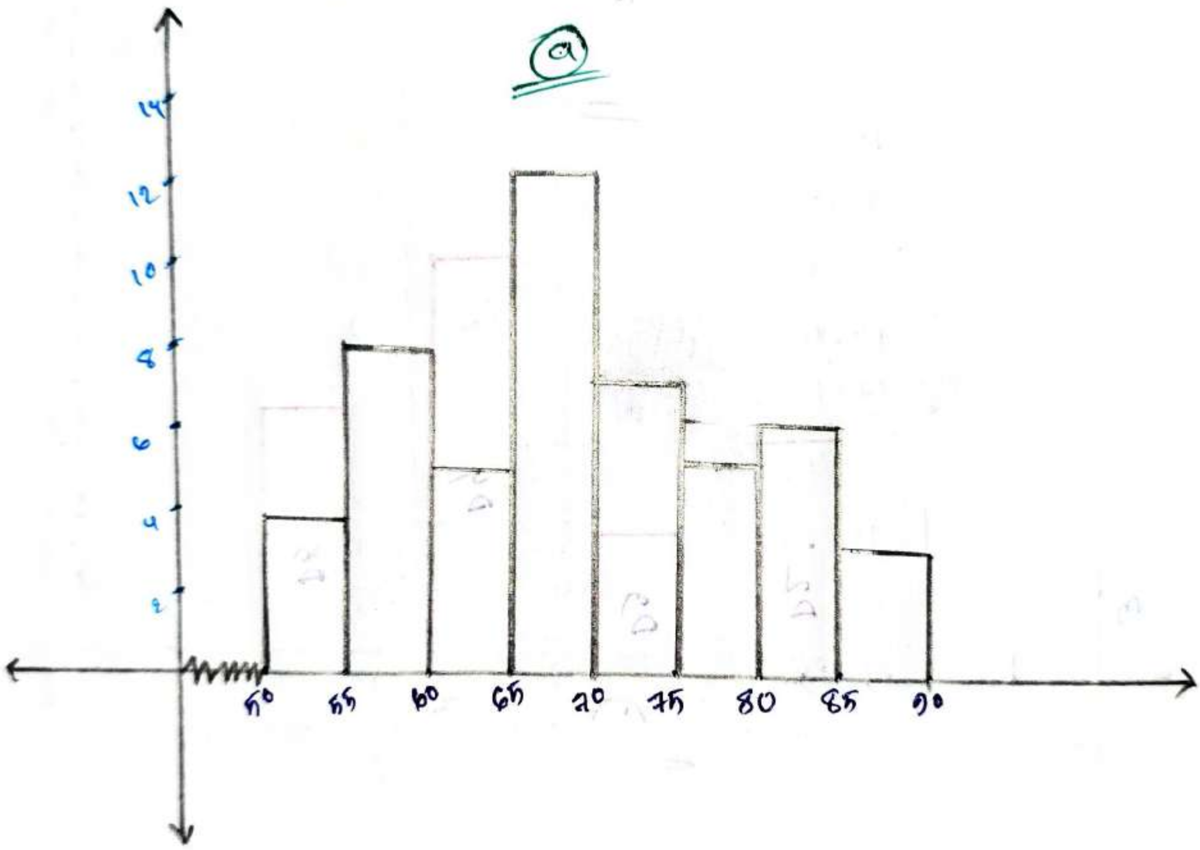


#problems 1.5:—

Weight	50-55	55-60	60-65	65-70	70-75	75-80	80-85	85-90	total
No.of Person	4	8	5	12	7	5	6	3	50

(a) Draw the histogram for the data.

(a)



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problem: 1.6

Pulse/min	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100	total
No. of Volunteers	2	7	11	15	10	9	6	3	63

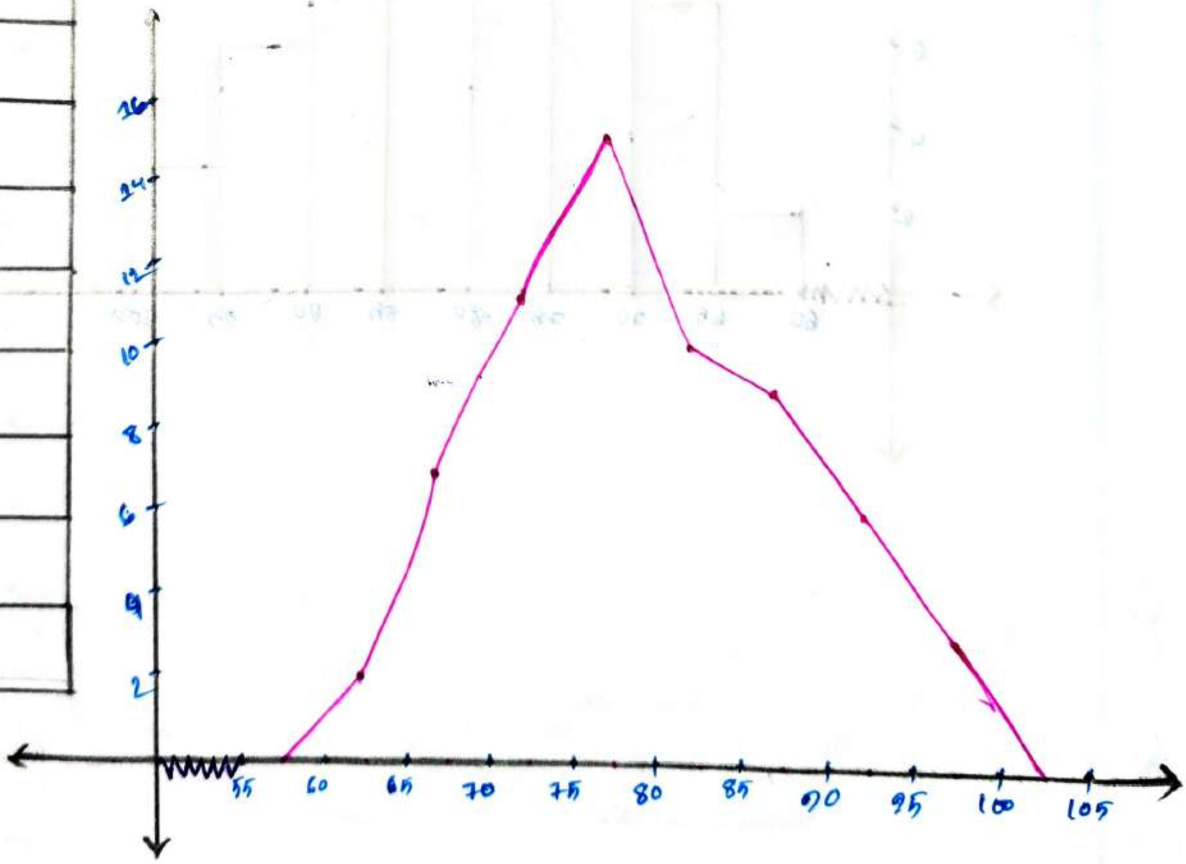
(a) Represent the data by histogram and frequency curve.

(a)

⇒ For Histogram:—

⇒ We have to need the midpoint of pulse/min.

Mid Value
62·5
67·5
72·5
77·5
82·5
87·5
92·5
97·5



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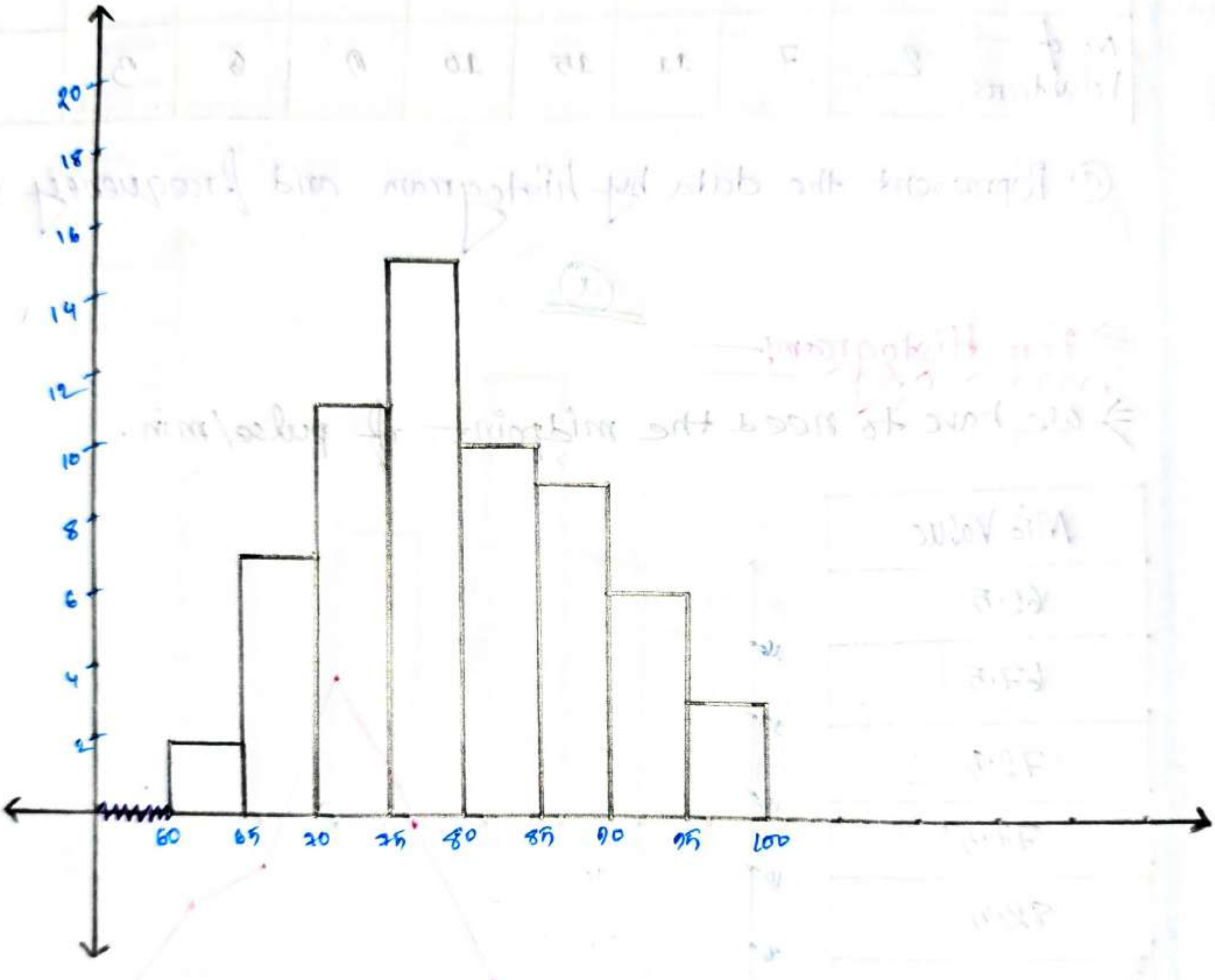
⇒ For Histogram:-



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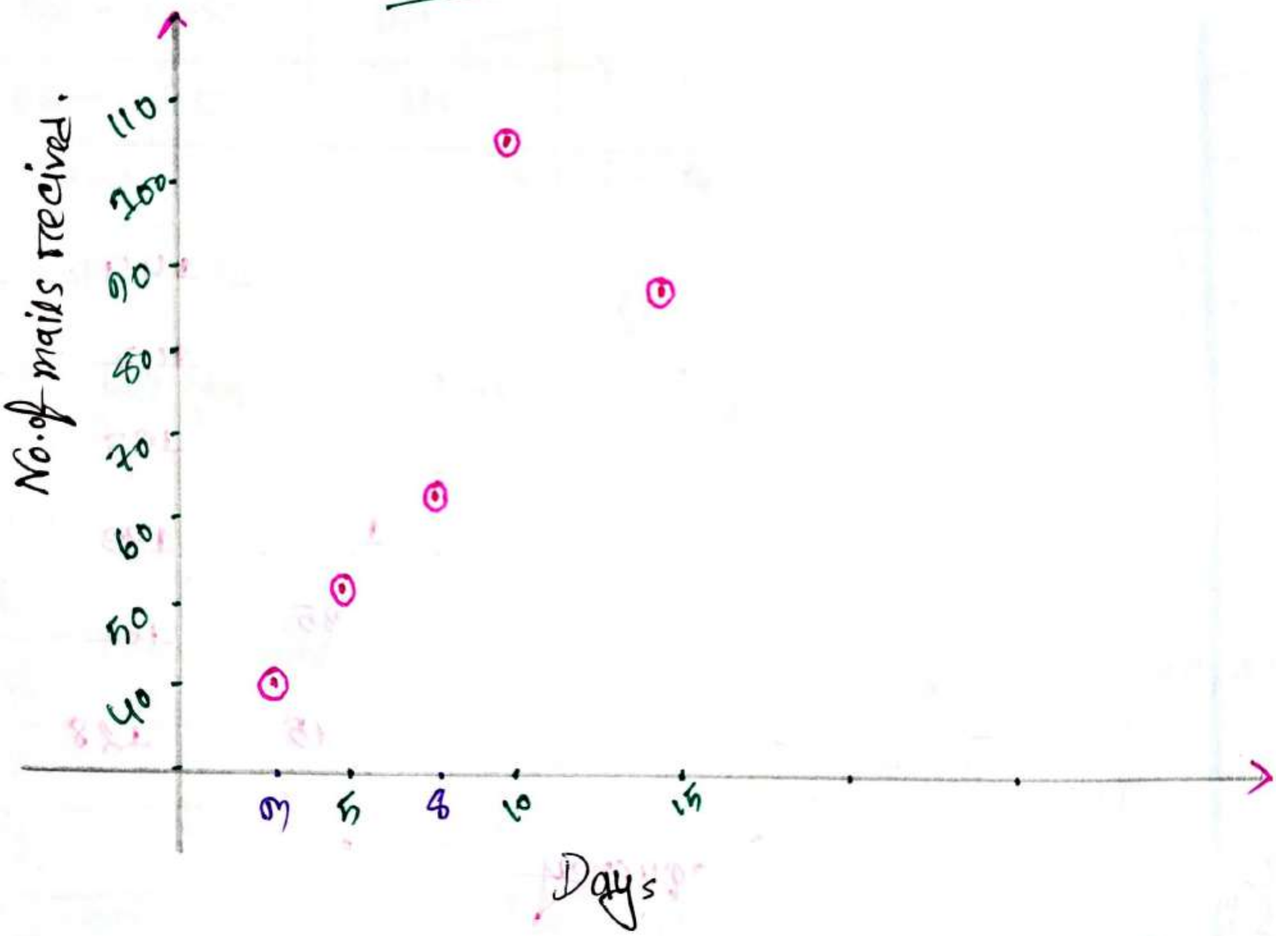
problem: 1.7:

Days	5	8	3	10	15
No. of mails received	54	65	42	107	87

(a). Draw the scatter Diagram of the data.

(a)



Tabulation Method:-

Some Rules from math:-

1. $[1,5) = 1 \le x < 5$
2. $(1,5) = 1 < x < 5$
3. $[1,5] = 1 \le x \le 5$
4. $(1,5] = 1 < x \le 5$

Example: 1.6:-

$\Rightarrow$

136	164	150	132	144	125	149
157	146	158	140	147	136	148
152	144	168	126	138	176	163
118	154	165	146	173	142	147
135	153	140	135	161	145	135
142	156	156	145	128		

(a) Construct a frequency distribution.

(b) Length < 158

(c) % < 158

(d) % which is above & 148

(a)

Class Size	Tally	Frequency	Cumulative Frequency
[118 - 128)	III	3	3
[128 - 138)	~~IIII~~ II	7	10
138 - 148	~~IIII~~ ~~IIII~~ III	13	23
148 - 158	~~IIII~~ IIII	9	32
158 - 168	~~IIII~~	5	37
168 - 178	III	3	40
Total		40	

(b)

⇒ Length are Less than 158 = 32

(c)

⇒ There are, $\frac{32}{40} \times 100 = 80\%$ Less than 158

(d)

⇒ There are, $\frac{9+5+3}{40} \times 100 = 42.5\%$ which is above & 148.

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Example: 2:—

pulse/min	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
student	2	7	11	15	10	9	6	3

ⓐ % of student with pulse less than 85

ⓑ % of student above & 85

ⓒ No. of student less than 90.

⇒ 1st Construct a frequency distribution.

pulse/min	Frequency	Cumulative Frequency
60 - 65	2	2
65 - 70	7	9
70 - 75	11	20
75 - 80	15	35
80 - 85	10	45
85 - 90	9	54
90 - 95	6	60
95 - 100	3	63
Total	63	

(a)

⇒ There are, $\frac{45}{63} \times 100 = 71.43\%$ less than 85.

(b)

⇒ There are, $\frac{9+6+3}{63} \times 100 = 28.57\%$ which is above ~~&~~ 85.

(c)

⇒ No. of student less than 90 = 54.

DESCRIPTIVE STATISTICS

Ch-2

Descriptive Statistics:-

⇒ It brief informational Coefficients that Summarize a given data set, which can be either a representation of the entire population or a sample of a population.

There are two types:-

1. Central tendency.
2. Dispersion.

Measure of Central Tendency:-

- Central Tendency / Average
  - Mathematical Average
    - Mean
      - Arithmetic Mean (AM)
      - Geometric Mean (GM)
      - Harmonic Mean (H.M)
  - Positional Average
    - Median
    - Mode

Formulas:—

	AM	GM	HM
Ungroup	$AM = \frac{1}{n}\sum_{i=1}^{n} x_i$ n = total no. of values.	$GM = \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$ $= \text{Antilog}\left[\frac{1}{n}\sum_{i=1}^{n}\log x_i\right]$	$HM = \frac{n}{\sum_{i=1}^{n}\frac{1}{x_i}}$
Group	$AM = \frac{1}{n}\sum_{i=1}^{n} f_i x_i$	$GM = \sqrt[n]{x_1^{f_1} \cdot x_2^{f_2} \cdots x_n^{f_n}}$ $= \text{Antilog}\left[\frac{1}{n}\sum_{i=1}^{n} f_i \log x_i\right]$	$HM = \frac{n}{\sum_{i=1}^{n} f_i/x_i}$
Advantage (A) & Disadvantage (D)	⇒ Not suitable for extreme values → D	⇒ Growth rate / Interest rate / ratio → A ⇒ 0 / –ve → D	⇒ wind, speed → A ⇒ 0 → D

Example 1:—

Prove that AM > GM > HM

⇒ Let, There are three Hypothetical values :— 1, 2, 3

We know:—

$$AM = \frac{1+2+3}{n} = \frac{6}{3} = 2$$

$n = 3$

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$$GM = \sqrt[n]{x_1 \cdot x_2 \cdot x_3}$$
$$= \sqrt[3]{1\times 2\times 3}$$
$$= (6\times 1)^{\frac{1}{3}}$$
$$= 1.81$$

$$H.M = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}}$$
$$= \frac{3}{\frac{1}{1}+\frac{1}{2}+\frac{1}{3}}$$
$$= 1.63$$

$\therefore$ $HM < GM < AM = 1.63 < 1.81 < 2$

Proved!

Example: 2!

$\Rightarrow$ Prove that $AM = GM = HM$

$\Rightarrow$ There are 3 Hypothetical values: 2, 2, 2

$\therefore$ we know!

$$AM = \frac{x_1+x_2+x_3}{n} = \frac{2+2+2}{3} = 2$$

$$GM = \sqrt[n]{x_1\times x_2\times x_3} = \sqrt[3]{2\times 2\times 2} = 2$$

$$HM = \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}} = \frac{3}{\frac{1}{2}+\frac{1}{2}+\frac{1}{2}} = 2$$

$\therefore$ $AM = GM = HM = 2$ proved

Example:- 1

⇒ The Quiz scores of 5 randomly selected Students are 15, 14, 13, 17, 15

(a) Find AM, GM, HM.

a

1st arrange the values:-

13, 14, 15, 15, 17

here,
$n = 5$

We know:-

For AM:-

$$AM = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{n}$$

$$= \frac{13+14+15+15+17}{5}$$

$$= 12.2 + 2.6$$

For GM:-

$$= 14.80$$

$$GM = \sqrt[n]{x_1 \times x_2 \times x_3 \times x_4 \times x_5}$$

$$= \sqrt[5]{13 \times 14 \times 15 \times 15 \times 17}$$

$$= 14.74$$

For HM:-

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} + \frac{1}{x_5}}$$

$$= \frac{5}{\frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{15} + \frac{1}{17}}$$

$$= 14.68$$

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For GM using Log:-

$$GM = \text{Antilog}\left[\frac{1}{n}\left\{\log(x_1 \times x_2 \times x_3 \times x_4 \times x_5)\right\}\right]$$

$$= \text{Antilog}\left[\frac{1}{5}\left\{\log(13 \times 15 \times 15 \times 14 \times 17)\right\}\right]$$

$$= 14.74$$

Ans:-

Example: 2:-

Class Interval	1-3	3-5	5-7	total		
No. of Stations	2	5	3	10		

(a) Calculate the AM, GM, HM.

(a)

⇒

Class	Frequency (f_i)	Mid Value (x_i)	$f_i x_i$	$f_i \log x_i$	f_i/x_i
1 - 3	2	2	4	2 log 2	1
3 - 5	4+1=5	4	20	5 log 4	1.25
5 - 7	6-3=03	6	18	3 log 6	0.5
Total	n = 10		$\Sigma f_i x_i = 42$	$\Sigma f_i \log x_i = 5.546$	$\Sigma f_i/x_i = 2.75$

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we know:—

For AM:—

$$AM = \frac{1}{n}\sum_{i=1}^{n} f_i x_i$$
$$= \frac{1}{10} \times \sum f_i x_i$$
$$= \frac{1}{10} \times 42$$
$$= 4.2$$

For GM:—

$$GM = \text{Antilog}\left[\frac{1}{n}\sum f_i \log x_i\right]$$
$$= \text{Antilog}\left[\frac{1}{10} \times 5.046\right]$$
$$= 3.03$$

For HM:—

$$HM = \frac{n}{\sum f_i/x_i}$$
$$= \frac{10}{2.75}$$
$$= 3.63$$

Exercise: 5

⇒ For Any two numbers. $AM = 10$ & $GM = 8$

(a) find out the numbers.

(a)

we know:—

$$AM = \frac{x_1 + x_2}{2} = 10$$
$$\Rightarrow x_1 + x_2 = 20 \quad \text{——— (i)}$$
$$GM = \sqrt[n]{x_1 \times x_2} = 8$$
$$\Rightarrow \sqrt[2]{x_1 \times x_2} = 8 \quad \text{——— (ii)}$$

is given

$GM = 8$

$AM = 10$

$n = 2$

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from eq (I) =)

$$x_1 = 20 - x_2 \quad \text{—— (III)}$$

put it into the eq (II) =)

$$(20-x_2)\cdot x_2 = 8^2$$

$$\Rightarrow 20x_2 - x_2^2 = 64$$

$$\Rightarrow x_2^2 - 20x_2 + 64 = 0$$

$$\therefore x_2 = 16, 4$$

Now, put the value of ~~eq~~ x_2 into the eq (III) =>

$$\therefore x_1 = 4, 16$$

Ans:-

Exercise: 6:-

=> A train moves 1st 80 km at speed 75 km/h, 2nd 70km at speed 85km/h, 3rd 85 km at speed 66 km/h and 4th 55km at speed 50 km/h. find the average speed throughout the journey.

=>

distance	80km	70km	85km	55km
speed	75 km/h	85km/h	66km/h	50km/h

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We know:-

For speed we have to ues H.M:-

Speed	Distance (f_i)	x_i
75	80	75
85	70	85
66	85	66
50	55	50

$$\therefore HM = \frac{80+70+85+55}{\frac{80}{75}+\frac{70}{85}+\frac{85}{66}+\frac{55}{50}}$$

$$= 67.7875.$$

We know:

$$HM = \frac{\text{total distance}}{\sum \frac{f_i}{x_i}}$$

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Positional Average:—

⇒ Formulas:—

	Median (M_e)	mode (M_0)
UnGroup	$M_e = \left\{\frac{1}{2}(n+1)\right\}^{th}$ obs → [n=odd] $= \frac{1}{2}\left[\frac{n}{2}^{th} \text{obs} + \left(\frac{n}{2}+1\right)^{th} \text{obs}\right]$ → [n=even]	
Group	$M_e = L + \frac{\frac{n}{2} - c}{f} \times h$ L = Lower limit of median class. h = Size of Median class. f = frequency of median class c = Cumulative frequency of previous class of median class Median class = that class where c.f ≥ $\frac{n}{2}$	$M_0 = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$ L = Lower Limit of M_0 class h = Size of M_0 class. f_m = frequency of M_0 class f_1 = frequency of previous M_0 class. f_2 = frequency of next M_0 class. M_0 class = that class where frequency is highest

GOOD LUCK

Example:1:—

The given following data refer to the profits of a store in thousand taka for the last 12 months are 3, 6, 8, 9, 6, 10, 5, 12, 9, 8, 11, 7 Compute the median profit of the store.

⇒ 1st Arrange the data:—

3, 5, 6, 6, 7, 8, 8, 9, 9, 10, 11, 12 → n = 12 = even

We know:—

$$Me = \frac{1}{2}\left[\frac{n}{2}^{th} obs + \left(\frac{n}{2}+1\right)^{th} obs\right]$$

$$= \frac{1}{2}\left[\frac{12}{2}^{th} obs + \left(\frac{12}{2}+1\right)^{th} obs\right]$$

$$= \frac{1}{2}\left[6^{th} obs + 7^{th} obs\right]$$

$$= \frac{1}{2}[8+8]$$

$$= 8$$

Ans:—

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Example : 2 :

⇒ The given following data are 3, 4, 2, 3, 4, 5, 4, 3, 5, 7, 12. Find the Median of the data set.

⇒ 1st arrange the data :

2, 3, 3, 3, 4, 4, 4, 5, 5, 7, 12

$n = 11 = \text{odd}$

We know :

$$Me = \left\{\frac{1}{2}(n+1)\right\}^{th} \text{ obs}$$

$$= \left\{\frac{1}{2}(11+1)\right\}^{th} \text{ obs}$$

$$= 6^{th} \text{ obs}$$

$$= 4$$

Ans :

Example: 2·8:—

No. of hours worked	30–55	55–80	80–105	105–130	130–155	155–180	180–205
No. of worker	3	4	6	9	12	11	5

ⓐ Compute Median.

ⓑ Compute mode.

⇒ Is given:—
& we know:—

No. of hours worked	Frequency, f	Cumulative frequency (cf)
30 – 55	3	3
55 – 80	4	7
80 – 105	6	13
105 – 130	9	22 → c
Me class → Mo class → 130 – 155	12	34 → $cf \geq \frac{n}{2}$
155 – 180	11	45
180 – 205	5	50
Total	n=50	

ⓐ

we know:—

$$Me = L + \frac{\frac{n}{2} \times 1 - c}{f} \times h$$

$$= 130 + \frac{25 - 22}{12} \times 25$$

$$= 136.25$$

$\frac{n}{2} = \frac{50}{2} = 25$

∴ Me class = 130–155

$Cf \geq \frac{n}{2}$

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(b)

We know:-

$$Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$= 130 + \frac{12-9}{2\times12-9-11} \times 25$$

$$= 148.75$$

Ans:-

Exercise:-

7. Find an appropriate measure of central tendency with justification of following observations.

obs (x): 185, −12, 16, −16, 0, 15

⇒ We know:-

For AM → Not suitable for extreme values. (185) ✗

For GM → Not suitable for 0/−ve values (0, −16) ✗

For HM → Not suitable for 0 values (0) ✗

For Mo → Suitable for repeat values (No repeat values) ✗

For Me → Suitable for handling extreme values ✓

Now Rearrange the data:-

−16, −12, 0, 15, 16, 185

$n = 6 =$ even

We know:-

$$Me = \frac{1}{2}\left[\left\{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2}+1\right)^{th}\right\} obs\right]$$

$$= \frac{1}{2}[3^{th} obs + 4^{th} obs] = 0.5\{0+15\} = 7.5$$ Ans:-

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(11)

Class Size	10-20	20-30	30-40	40-50	50-60	total
No. of Localities	3	5	9	3	2	22

⇒ Following data represent the number of Computers Centers in different localities in Dhaka. Calculate the Average Computer per localities.

⇒

Class Size	No. of localities (f)	Comulative frequency (cf)
10-20	3	3
20-30	5	8 → c
me.class, mo.class → 30-40	9	17
40-50	3	20
50-60	2	22
total	n = 22	

Using Median:-

We know:-

$\therefore Me = L + \frac{n/2 - c}{f} \times h$

$= 30 + \frac{11 - 8}{9} \times 10$

$= 33.33$

$\frac{n}{2} = 11$

Me.class (30-40)

$cf \geq \frac{n}{2}$

Ans:-

Using Mode:-

We know:-

$Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$

$= 30 + \frac{9 - 5}{2 \times 9 - 5 - 3} \times 10$

$= 34$

Ans:-

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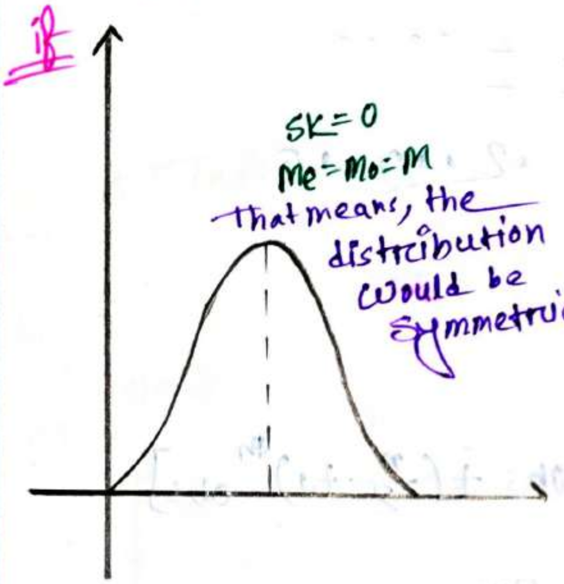
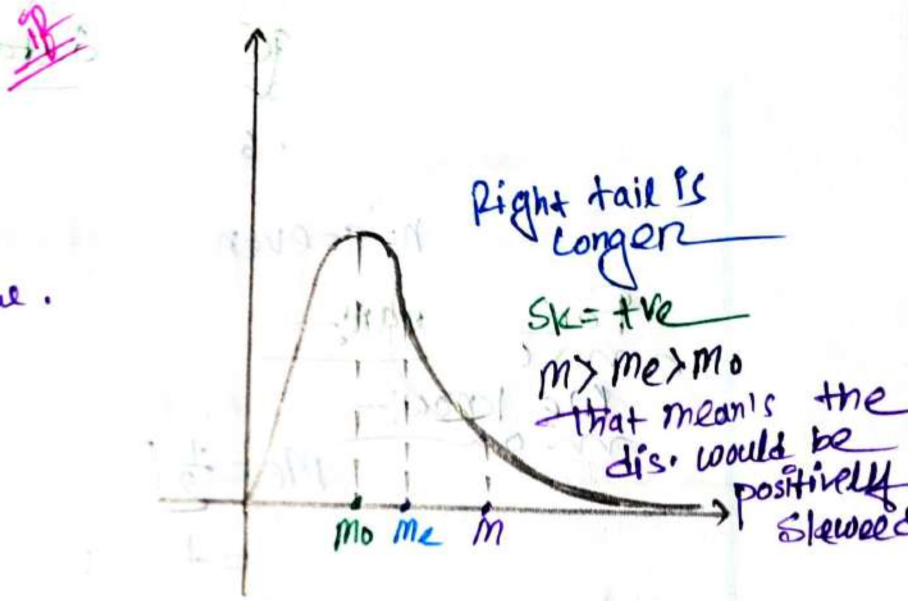
Skewness:—

⇒ It measure the lack of symmetry in data distribution.

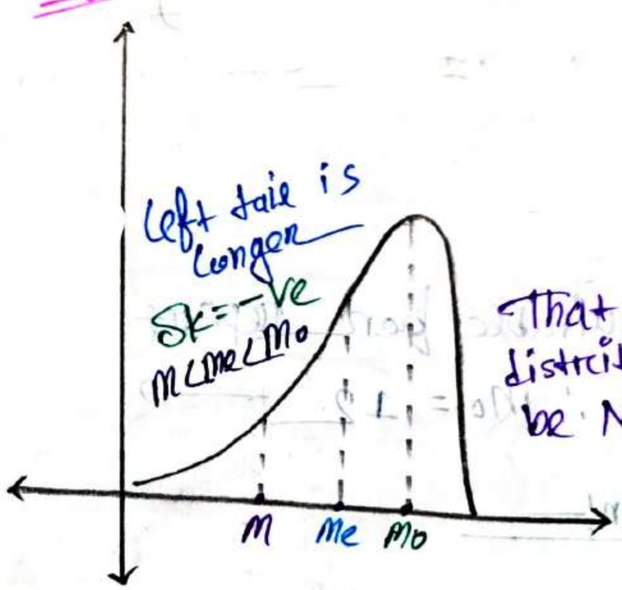
If



If



If:



Formula:—

$\therefore SK = M - Me$

OR, $SK = M - Mo$

Example:-1:-

The given following Data are, 10, 15, 12, 13, 12, 5, 7, 6 find the median & mode of the data. Comment of the Skewness of the given distributions.

⇒ 1st Rearrange the data:-

5, 6, 7, 10, 12, 12, 13, 15

$n = 8 = even$

For median:-

We know:-

$$Me = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} obs + \left(\frac{n}{2}+1\right)^{th} obs\right]$$

$$= \frac{1}{2}[4th + 5th]$$

$$= \frac{1}{2}[10 + 12]$$

$$= 11$$

For Mode:-

We know:-

Mo, Suitable for ~~repeat~~ Values

$\therefore Mo = 12$ → since it's twice

For Mean:-

we know:-

$$AM = \frac{1}{n}\left(\sum_{i=1}^{8} x_i\right)$$

$$= \frac{1}{8}(5+6+7+10+12+12+13+15)$$

$$= 10$$

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Now, we know:-

$$SK = AM - Me$$
$$= 10 - 11$$
$$= -1$$
$$= -ve$$

$$SK = AM - Mo$$
$$= 10 - 12$$
$$= -2$$
$$= -ve$$

∴ That mean's the distribution would be Negatively Skewed. → ***

Example: 2:-

Class size	10-20	20-30	30-40	40-50	50-60
No. of Localities	3	5	9	3	2

⇒ Comment of Skewness of the given distributions.

⇒

Class Size	frequency (f_i)	Mid value (x_i)	cf	$f_i x_i$
10-20	3	15	3	45
20-30	5	25	8→c	125
Me.class → Mo.class → 30-40	9	35	17	315
40-50	3	45	20	135
50-60	2	55	22	110
total	n= 22			$\sum f_i x_i = 730$

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For Mean:—

$$AM = \frac{\sum f_i x_i}{n}$$

$$= \frac{730}{22}$$

$$= 33.18$$

For Median:—

$$Me = L + \frac{n/2 - c}{f} \times h$$

$$= 30 + \frac{11-8}{9} \times 10$$

$$= 33.33$$

$\frac{n}{2} = 11$

$\therefore$ Me. class (30–40)

$\because cf \geq \frac{n}{2}$

For Mode:—

$$Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$= 30 + \frac{9-5}{2\times 9 - 5 - 3} \times 10$$

$$= 34$$

$\therefore$ Now we know:—

$$Sk = AM - Me$$

$$= 33.18 - 33.33$$

$$= -0.15$$

$$= -ve$$

$$Sk = AM - Mo$$

$$= 33.18 - 34$$

$$= -0.82$$

$$= -ve$$

That mean's the distribution would be Negatively Skewed

Ans.

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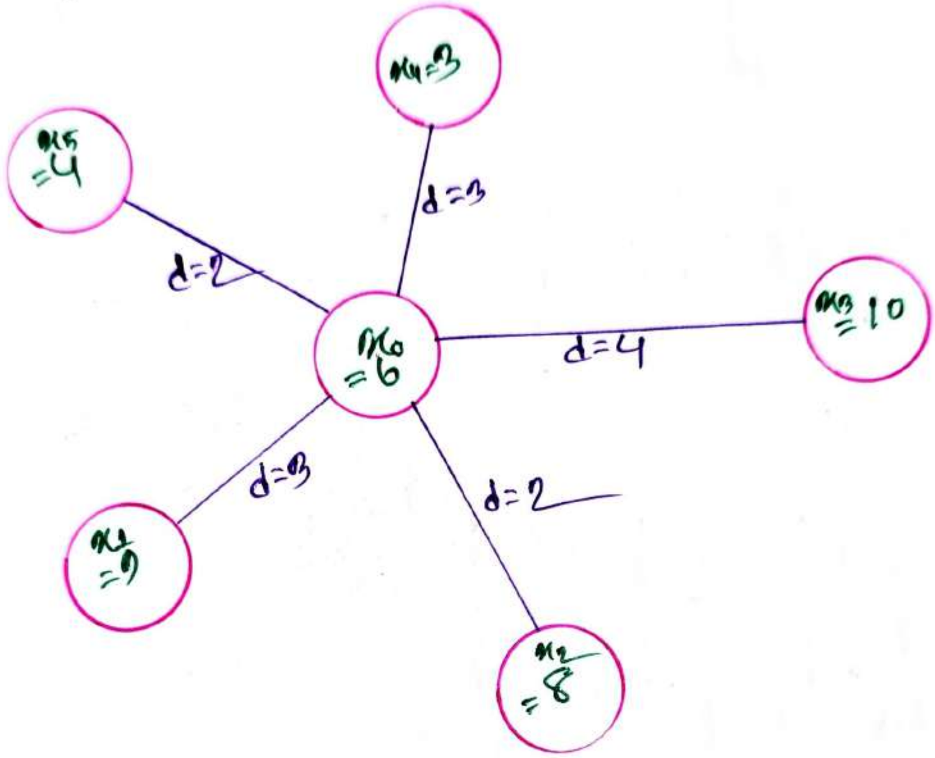
Dispersion:-

⇒ Despersion in a way to describing how spread Out a set of data.



Measures of dispersion:-

⇒ It discribing the shape of the data & spread of the data set.

There are 5 (Five) types:-

1. Mean deviation.
2. Variance.
3. Standard deviation.
4. Coefficient of Variation.
5. Quartiles.

Formulas:—

	(MD) Mean Deviation	Variance (σ^2)
UnGroup	$MD = \frac{1}{n}\sum \lvert x_i - \bar{x} \rvert$	$\sigma^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$
Group	$MD = \frac{1}{n}\sum f_i \lvert x_i - \bar{x} \rvert$	$\sigma^2 = \frac{1}{n}\sum f_i (x_i - \bar{x})^2$
	Standard Deviation (SD)(σ)	(CV) Co. eff. of Variation.
	$SD = \sqrt{\sigma^2}$	$CV = \frac{SD}{\bar{x}} \times 100\%$ OR $\Rightarrow \frac{\sigma}{\bar{x}} \times 100\%$

For Ungroup:—

$$\Rightarrow \bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + \dots + x_n}{n} = \sum_{i=1}^{n} x_i / n$$

For Group:—

$$\Rightarrow \bar{x} = \frac{\sum f_i x_i}{n}$$

problem: 1-

⇒ The following data set : 2, 4, 8, 6, 10, 12

Compute,

(a) MD

(b) σ^2

(c) SD

(d) CV

⇒ In given:-

$x: 2, 4, 8, 6, 10, 12$

We know-

$$\bar{x} = \frac{x_1+x_2+x_3+x_4+x_5+x_6}{n} \quad \Bigg| \quad n=6$$

$$= \frac{2+4+8+6+10+12}{6}$$

$$= 7$$

$\lvert x_i-\bar{x}\rvert$	$(x_i-\bar{x})^2$
5	25
3	9
1	1
1	1
3	9
5	25
$\Sigma\lvert x_i-\bar{x}\rvert = 18$	$\Sigma(x_i-\bar{x})^2 = 70$

(a)

we know:-

$$MD = \frac{1}{n}\Sigma\lvert x_i-\bar{x}\rvert = \frac{1}{6}\times 18 = 3$$

(b)

We know:-

$$\sigma^2 = \frac{1}{n}\Sigma(x_i-\bar{x})^2 = \frac{1}{6}\times 70 = 11.67$$

(c)

we know:-

$$SD = \sqrt{\sigma^2} = \sqrt{(11.67)^2} = 3.41$$

(d)

we know:-

$$CV = \frac{SD}{\bar{X}}\times 100\% = \frac{3.41}{7}\times 100\% = 48.79\%$$

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Exercise:-

10:- A Company has two section with $\overset{A}{40}$ and $\overset{B}{65}$ staffs respectively. Their Average weekly wages are \$450 and \$350. the standard deviations are 7 and 9. Which section has larger variability in wages?

⇒ We know:-

$$CV = \frac{SD}{\bar{x}} \times 100\%$$

$\therefore CV_A = \frac{SD_A}{\bar{X}_A} \times 100\%$ | $SD_A = 7$, $\bar{x}_A = 450$

$= \frac{7}{450} \times 100\%$

$= 1.56\%$

$\therefore CV_B = \frac{SD_B}{\bar{X}_B} \times 100\%$ | $SD_B = 9$, $\bar{X}_B = 350$

$= \frac{9}{350} \times 100\%$

$= 2.57\%$

∴ Section B has more Variability in Wages.

10.1 A Company has two Section with $\overset{A}{40}$ And $\overset{B}{65}$ staffs respectively. their total Monthly wages are \$150000 & \$14100. The standard deviation are 11 and 17. Which Section has larger Variability in wages?

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⇒ We know:-

$$CV = \frac{SD}{\bar{x}} \times 100\% \quad \text{——— (I)}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \quad \text{——— (II)}$$

here,
$\sum x_{iA} = 150000$
$n_A = 40$
$\sum x_{iB} = 141000$
$n_B = 65$
$SD_A = 11$
$SD_B = 17$

$$\therefore \bar{x}_A = \frac{150000}{40} = 3750$$

$$\bar{x}_B = \frac{141000}{65} = 2169.23$$

From eq (I) ⇒

$$CV_A = \frac{SD_A}{\bar{x}_A} \cancel{\times 100\%} = \frac{11}{3750} \times 100\% = 0.293\%$$

$$CV_B = \frac{SD_B}{\bar{x}_B} \times 100\% = \frac{17}{2169.23} \times 100\% = 0.783\%$$

∴ Section B has more Variability in wages.

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(1) No. of employees of a multinational Company of 10 Countries are 44, 50, 38, 96, 42, 47, 40, 39, 46, 50.

(a) Find MD

(b) σ^2

(c) SD

(d) CV

$\Rightarrow$ We know:

$\bar{x} = \sum x_i / n$ —— (1)

x_i	$(x_i-\bar{x})^2$	$\lvert x_i-\bar{x}\rvert$
44	27.04	5.2
50	0.64	0.8
38	125.44	11.2
96	2190.24	46.8
42	51.84	7.2
47	4.84	2.2
40	84.64	9.2
39	104.04	10.2
46	10.24	3.2
50	0.64	0.8
$\sum x_i = 492$	$\sum (x_i-\bar{x})^2 = 2599.6$	$\sum \lvert x_i-\bar{x}\rvert = 96.8$

From eq (1) $\Rightarrow$

$\bar{x} = \frac{492}{10}$ $\left[n = 10 \right.$

$= 49.2$

(a)

We know:-

$$MD = \frac{1}{n}\sum|x_i-\bar{x}| = \frac{1}{10}\times 96.8 = 9.68$$

(b)

We know:-

$$\sigma^2 = \frac{1}{n}\sum(x_i-\bar{x})^2 = \frac{1}{10}\times 2596.6 = 259.66$$

(c)

We know:-

$$SD = \sqrt{\sigma^2} = \sqrt{259.66} = 16.11$$

(d)

We know:-

$$CV = \frac{SD}{\bar{x}}\times 100\% = \frac{16.11}{49.2}\times 100\% = 32.74\%$$

Ans:-

Example:- 2.12:-

Overtime	10−15	15−20	20−25	25−30	30−35
No. of Employees	3	5	7	4	2

Find:- (a) MD
(b) σ^2
(c) SD
(d) CV

⇒

overtime	mid value x_i	f_i	$f_i x_i$	$\lvert x_i-\bar{x}\rvert$	$(x_i-\bar{x})^2$
10−15	12.5	3	37.5	9.285	86.211
15−20	17.5	5	87.5	4.285	18.361
20−25	22.5	7	157.5	0.715	0.511
25−30	27.5	4	110	5.715	32.661
30−35	32.5	2	65	10.715	114.811
		$\Sigma n=21$	$\Sigma f_i x_i=457.5$	$\Sigma\lvert x_i-\bar{x}\rvert=30.715$	$\Sigma(x_i-\bar{x})^2=252.555$

⇒ here,

$n=21$

$$\therefore \bar{x}=\frac{\Sigma f_i x_i}{n}=\frac{475.5}{21}=21.785$$

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$f_i\lvert x_i-\bar{x}\rvert$	$f_i(x_i-\bar{x})^2$
27.855	258.633
21.425	91.806
5.505	3.578
22.860	130.644
21.430	229.622
$\sum f_i\lvert x_i-\bar{x}\rvert = 98.575$	$\sum f_i(x_i-\bar{x})^2 = 714.29$

(a)

We know:—

$$MD = \frac{1}{n}\sum f_i\lvert x_i-\bar{x}\rvert = \frac{1}{21}\times 98.57 = 4.69$$

(b)

We know:—

$$\sigma^2 = \frac{1}{n}\sum f_i(x_i-\bar{x})^2 = \frac{1}{21}\times 714.29 = 34.01$$

(c)

We know:—

$$\sigma = \sqrt{\sigma^2} = \sqrt{34.01} = 5.832$$

(d)

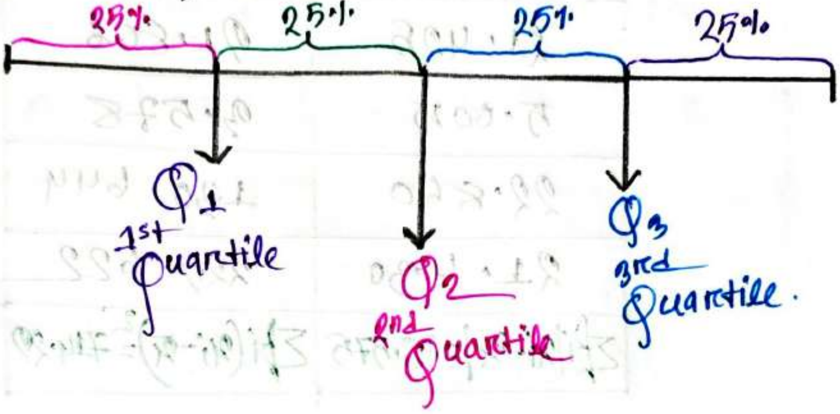
We know:—

$$CV = \frac{SD}{\bar{x}}\times 100\% = \frac{5.832}{21.785}\times 100\% = 26.77\%$$

Ans:—

Quartiles:-

⇒ There are the values that break down the dataset into quarters, or quartiles.



	UnGroup	Group
n=odd	$Q_i = \left[i \times \frac{n+1}{4}\right]^{th}$ obs	$Q_i = L + \frac{\frac{in}{4} - c}{f} \times h$ L = Lower Limit of Qur. class h = Size of Qur. class. f = frequency of Qur. class c = Cumulative frequency of previous class
n=even	$Q_i = \frac{1}{2}\left[\frac{i(n)}{4}^{th}\text{ obs} + \left(\frac{in}{4}+1\right)^{th}\text{obs}\right]$	

Note:-

⇒ Quartile class in that class where cumulative frequency $cf \geq \frac{in}{4}$

Example:- 2.17

⇒ A data set of the no. of OPD patients in a clinic in a week is given as,

$x: 2, 5, 3, 6, 7, 4, 9$

Find:-

(a) Q_1, Q_2, Q_3

⇒ 1st re arrange the data:-

$x: 2, 3, 4, 5, 6, 7, 9$

$n = 7 = \text{odd}$

(a)

For Q_1:-

We know:- $Q_i = \left[\frac{i(n+1)}{4}\right]^{th}$ obs

$\therefore Q_1 = \left\{\frac{1\times(7+1)}{4}\right\}^{th}$ obs

$= 2^{th}$ obs

$= 3$

For Q_2:-

$Q_2 = \left\{\frac{2\times(7+1)}{4}\right\}^{th}$ obs

$= 4^{th}$ obs

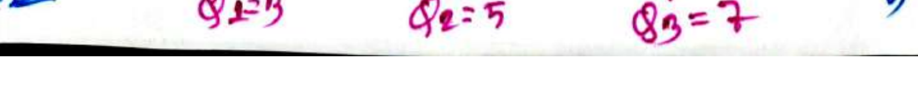
$= 5$

For Q_3:-

$Q_3 = \left\{\frac{3\times(7+1)}{4}\right\}^{th}$ obs

$= 6^{th}$ obs

$= 7$



Example:- 2.17.2:-

A given data set is:-

$x: 2, 5, 3, 6, 7, 4, 9, 13$

(a) Solve Q_1, Q_2, Q_3

$\Rightarrow$ 1st Rearrange the data:-

$x: 2, 3, 4, 5, 6, 7, 9, 13$

$n = 8$

We know:-

$$Q_i = \frac{1}{2}\left[\frac{in}{4}^{th} \text{ obs} + \left(\frac{in}{4}+1\right)^{th} \text{obs}\right]$$

For Q_1:-

$$Q_1 = \frac{1}{2}\left[\frac{1\times 8}{4}^{th} \text{ obs} + \left(\frac{1\times 8}{4}+1\right)^{th} \text{ obs}\right]$$

$$= \frac{1}{2}[\text{2th obs} + \text{3rd obs}]$$

$$= \frac{1}{2}[3+4] = 3.5$$

For Q_2:-

$$Q_2 = \frac{1}{2}\left[\frac{2\times 8}{4}^{th} \text{ obs} + \left(\frac{2\times 8}{4}+1\right)^{th} \text{ obs}\right]$$

$$= \frac{1}{2}[\text{4th obs} + \text{5th obs}]$$

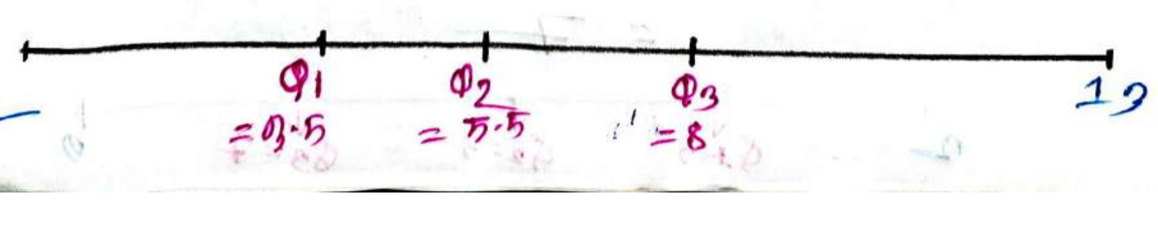
$$= \frac{1}{2}[5+6] = \frac{1}{2}[11] = 5.5$$

For Q_3:-

$$Q_3 = \frac{1}{2}\left[\frac{3\times 8}{4}^{th} \text{ obs} + \left(\frac{3\times 8}{4}+1\right)^{th} \text{obs}\right]$$

$$= \frac{1}{2}(\text{6th obs} + \text{7th obs})$$

$$= \frac{1}{2}(7+9) = 8$$



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Examples 2.10:-

Temp.	0-10	10-20	20-30	30-40	40-50	50-60
No of cities	2	3	5	2	6	2

Q:- Find Q_1, Q_2, Q_3

⇒ 1st find the cumulative frequency:-

Temp	Frequency	Cumulative frequency (cf)
0 - 10	2	2 → c_1 for Q_1
10 - 20 (Qur. class for Q_1)	3	5 → c_2 for Q_2
20 - 30 (Qur. class for Q_2)	5	10
30 - 40	2	12 → c_3 for Q_3
40 - 50 (Qur. class for Q_3)	6	18
50 - 60	2	20
total	n=20	

∴ we know:-

$$Q_i = l + \frac{\frac{in}{4} \times 1 - c}{f} \times h \quad ; \; cf \geq \frac{in}{4}$$

For Q_1:-

$$cf \geq \frac{1 \times 20}{4} = 5$$

∴ Qur. class = (10-20)

$$\therefore Q_1 = 10 + \frac{5-2}{3} \times 10$$

$$= 20$$

For : Q_2 :—

$$cf \geq \frac{2 \times 20}{4} = 10$$

$\therefore$ Qur. class = (20 − 30)

$$\therefore Q_2 = 20 + \frac{10-5}{5} \times 10$$

$$= 30$$

For Q_3 :—

$$cf \geq \frac{3 \times 20}{4} = 15$$

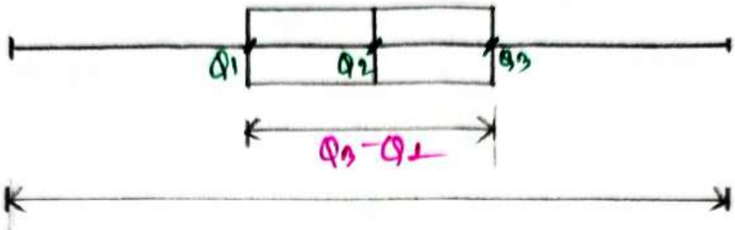
$\therefore$ Qur. class = (40 − 50)

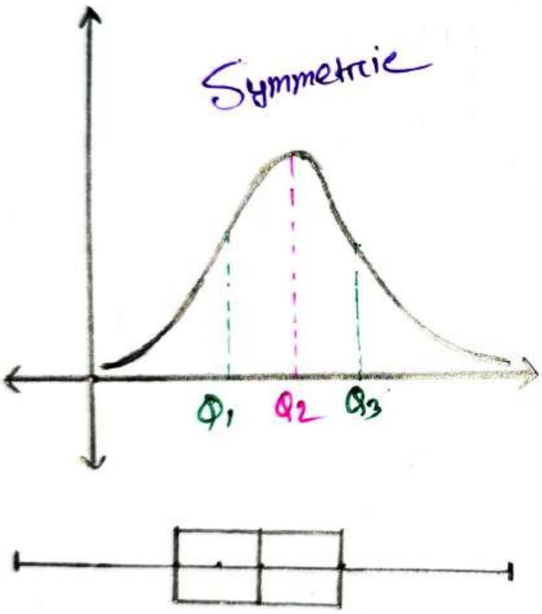
$$\therefore Q_3 = 40 + \frac{15-12}{6} \times 10$$

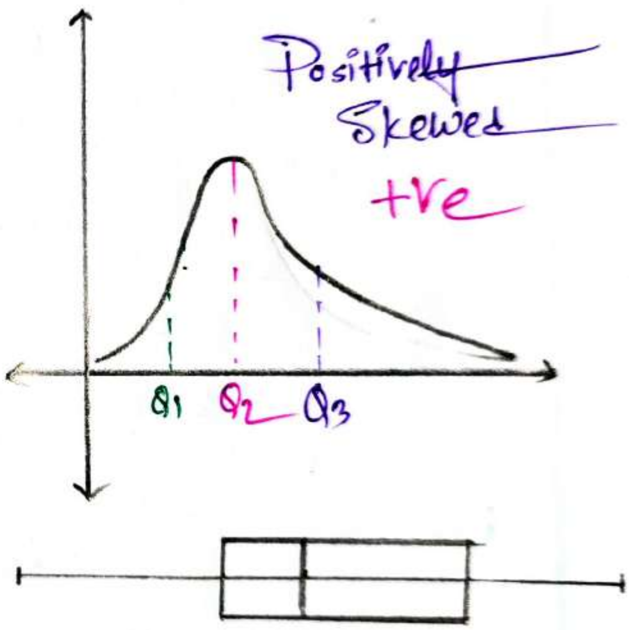
$$= 45$$

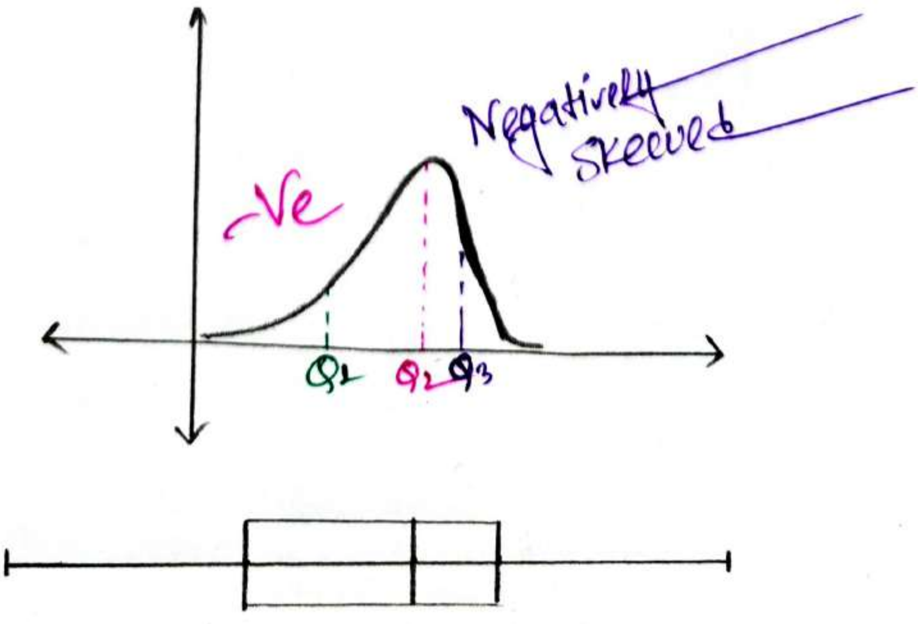
Box & whisker plot

⇒ Box-plots are a standardized way of displaying the distribution of data based on 5-number.









Example:- 2.17

⇒ A data set of the no. of OPD patients in a clinic in a week is given as,

x: 2, 5, 3, 6, 7, 4, 9

Find:-

(a) Q_1, Q_2, Q_3 Also draw the Box & Whisker plot

⇒ 1st re arrange the data:-

x: 2, 3, 4, 5, 6, 7, 9

$n = 7 =$ odd

(a)

For Q_1:-

We know:- $Q_i = \left[\frac{i(n+1)}{4}\right]^{th}$ obs

$\therefore Q_1 = \left\{\frac{1\times(7+1)}{4}\right\}^{th}$ obs

$= 2^{th}$ obs

$= 3$

For Q_2:-

$Q_2 = \left\{\frac{2\times(7+1)}{4}\right\}^{th}$ obs

$= 4th$ obs

$= 5$

For Q_3:-

$Q_3 = \left\{\frac{3\times(7+1)}{4}\right\}^{th}$ obs

$= 6th$ obs

$= 7$

Example:- 2.17.2:-

A given data set is:-

$x:$ 2, 5, 3, 6, 7, 4, 9, 13

ⓐ Solve Q_1, Q_2, Q_3 & Also draw the Box-and-whisker plot

$\Rightarrow$ 1st Rearrange the data:-

$x:$ 2, 3, 4, 5, 6, 7, 9, 13

$n = 8$

We know:-

ⓐ

$$Q_i = \frac{1}{2}\left[\frac{in}{4}^{th} \text{ obs} + \left(\frac{in}{4}+1\right)^{th} \text{obs}\right]$$

For Q_1:-

$$Q_1 = \frac{1}{2}\left[\frac{1\times 8}{4}^{th} \text{ obs} + \left(\frac{1\times 8}{4}+1\right)^{th} \text{ obs}\right]$$

$$= \frac{1}{2}[\text{2th obs} + \text{3rd obs}]$$

$$= \frac{1}{2}[3+4] = 3.5$$

For Q_2:-

$$Q_2 = \frac{1}{2}\left[\frac{2\times 8}{4}^{th} \text{ obs} + \left(\frac{2\times 8}{4}+1\right)^{th} \text{ obs}\right]$$

$$= \frac{1}{2}[\text{4th obs} + \text{5th obs}]$$

$$= \frac{1}{2}[5+6] = \frac{1}{2}(11) = 5.5$$

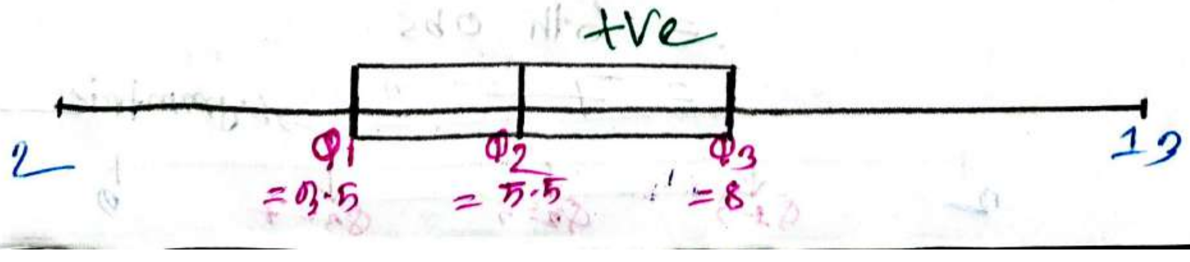
For Q_3:-

$$Q_3 = \frac{1}{2}\left[\frac{3\times 8}{4}^{th} \text{ obs} + \left(\frac{3\times 8}{4}+1\right)^{th} \text{obs}\right]$$

$$= \frac{1}{2}(\text{6th obs} + \text{7th obs})$$

$$= \frac{1}{2}(7+9) = 8$$

+ve



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Examples 2.10:-

Temp.	0-10	10-20	20-30	30-40	40-50	50-60
No of cities	2	3	5	2	6	2

Q:- Find Q_1, Q_2, Q_3 & Also Draw a box & wshiker plot

⇒ 1st find the Cumulative frequency:-

Temp	Frequency	Cumulative frequency (cf)
0 - 10	2	2 → c_1 for Q_1
Qur class for Q_1 → 10 - 20	3	5 → c_2 for Q_2
Qur. class for Q_2 → 20 - 30	5	10
30 - 40	2	12 → c_3 for Q_3
Qur class for Q_3 → 40 - 50	6	18
50 - 60	2	20
total	n=20	

∴ We know:-

$$Q_i = l + \frac{\frac{in}{4} \times 1 - c}{f} \times h \quad ; \; cf \geq \frac{in}{4}$$

For Q_1:-

$$cf \geq \frac{1 \times 20}{4} = 5$$

∴ Qur. class = (10-20)

$$\therefore Q_1 = 10 + \frac{5-2}{3} \times 10$$

$$= 20$$

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For Q2:—

$$cf \geq \frac{2\times 20}{4} = 10$$

∴ Qur. class = (20 − 30)

$$\therefore Q_2 = 20 + \frac{10-5}{5}\times 10$$

$$= 30$$

For Q3:—

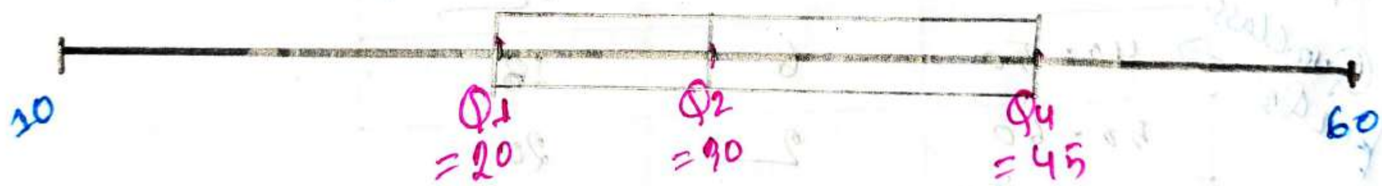
$$cf \geq \frac{3\times 20}{4} = 15$$

∴ Qur. class = (40 − 50)

$$\therefore Q_3 = 40 + \frac{15-12}{6}\times 10$$

$$= 45$$

+ve



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PROBABILITY

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Probability:-

⇒ It is the measure of the likelihood that an event will occur. It is quantified as a number between 0 & 1.

☑ Some relevent terms which are needed to Understand

⇒ Experiment:-

→ An Activity or process where the outcome is uncertain, like, tossing a coin or rolling a dice.

⇒ OutCome:-

→ A Single possible result of an experiment.

There are two types:-

1. Favorable → No. of outcome in favor of an event

2. Exhaustive → All possible outcome of an random exp. like for Sample space (S) given above the outcomes HH, HT, TH, TT

⇒ Sample Space:-

→ The Collection of all possible outcomes of an exprement.

Ex:- Tossing a Coin twice {Outcome: H, T} {Sample space: S = {HH, HT, TH, TT}}

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⇒ Event:—

→ A specific outcome or set of outcomes within a sample space.

They are 4 types:—

1. Independent → Two or more events are said to be independent when the occurrence of one trial does not affect the other.
   Ex:- A coin tossed twice.

2. Equally likely :→ When every Outcome in a sample space has the same probability of Occurring.

3. Mutually exclusive:→ When they don't Occur Simultaneously.

4. Complementary:— If the probability of an event A, P(A) then probability of that event would not occur is 1 - P(A)

⇒ probability:—

$$P(A) = \frac{\text{No. of Favourable outcomes}}{\text{Total possible outcome}}$$

$$\therefore \boxed{P(A) = \frac{n(A)}{n(S)}}$$

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⇒ Conditional probability :—

$$\boxed{P(A/B) = \frac{P(A \cap B)}{P(B)}}$$

→ It is the probability that event A occurs given that event B has already occurred. denote as $P(A/B)$. Read as the probability of A given B.

⇒ Additive law of probability :—

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

⇒ Multiplicative law of probability :—

$$\boxed{P(AB) = P(A) \cdot P(B)}$$

Where,

A, B are independent.

Examples 3·3:—

⇒ three fair Coins tossed once. What is the probability that.

ⓐ At least two coin will show head.

ⓑ 1st & 3rd coin will show head.

ⓒ 1st coin show head given that 3rd coin show head.

⇒ probability tree of three fair Coins tossed:—

$\therefore S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

ⓐ

At Least two coin show head $A = \{HHH, HHT, HTH, THH\}$

We know:—

$$P(A) = \frac{n(A)}{n(S)}$$

$$= \frac{4}{8}$$

$$= \frac{1}{2}$$

(b)

1st & 3rd Coin show head $B = \{HHH, HTH,\}$

We know:-

$$P(B) = \frac{n(B)}{n(S)}$$

$$= \frac{2}{8} = \frac{1}{4}$$

(c)

1st Coin show head $A = \{HHH, HHT, HTH, HTT\}$

3rd coin show head $B = \{HHH, HTH, THH, TTH\}$

$\therefore (A \cap B) = \{HHH, HTH\}$

We know:-

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{\frac{2}{8}}{\frac{4}{8}} = \frac{2}{4} = \frac{1}{2}$$

Ans:-

(d) 1st or 3rd coin will show head

(d)

$\Rightarrow$ From c:-

$$P(A) = \frac{4}{8}$$

$$P(B) = \frac{4}{8}$$

$$P(A \cap B) = \frac{2}{8}$$

We know:-

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{4}{8} + \frac{4}{8} - \frac{2}{8} = \frac{6}{8}$$

Examples 3.7:—

⇒ In a box there are 30 bulbs. The bulbs are identified by identity number 1 to 30. One bulb is selected randomly. Find the probability that the selected bulb

(a) either multiple of 3 or 5.

(b) even, under the condition that it is multiple of 3.

(a)

⇒ Multiple of 3:—

$$A = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30\}$$

Multiple of 5:—

$$B = \{5, 10, 15, 20, 25, 30\}$$

$$\therefore A \cap B = \{15, 30\}$$

We know:—

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{10}{30} + \frac{6}{30} - \frac{2}{30}$$

$$= \frac{7}{15}$$

Ans:—

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(b)

From a:-

multiple of 3:-

$P(A) = \frac{10}{30}$

Even Number:-

$C = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 28, 26, 30\}$

$\therefore P(C) = \frac{15}{30}$

$\therefore (A \cap C) = \{6, 12, 18, 24, 30\}$

We know:-

$$P(C/A) = \frac{P(A \cap C)}{P(A)}$$

$$= \frac{\frac{5}{30}}{\frac{10}{30}}$$

$$= \frac{1}{2}$$

Ans:-

Example: 30:-

⇒ Three Consecutive phone calls are monitored. The call maybe either Voice call (V). or Data call (D). where $P(V) = P(D) = 50\%$

Find the probability ~~that~~.

(a) no data call.

(b) At least one voice call.

(c) At best one voice call.

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⇒ The Sample Space:-

$$S = \begin{Bmatrix} VVV, VVD, VDV, VDD \\ DVV, DVD, DDV, DDD \end{Bmatrix}$$

(a)

$\therefore P(\text{no data calls}) = P(VVV) = \frac{1}{8}$

(b)

$\therefore P(\text{at least one Voice call}) = 1 - P(DDD)$

$= 1 - \frac{1}{8}$

$= \frac{7}{8}$

(c)

$\therefore P(\text{at best one Voice call}) = P(VDD, DVD, DDV, DDD)$

$= \frac{4}{8}$

Ans:-

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Example 3.10:-

⇒ Repeat Example 3.9; where $P(D) = 90\%$ & $P(V) = 10\%$

⇒ The Sample Space:-

$$S = \left\{ \begin{matrix} VVV, & VVD, & VDV, & VDD \\ DVV, & DVD, & DDV, & DDD \end{matrix} \right\}$$

(a)

$\therefore P(\text{No data calls}) = P(VVV) = P(V) \times P(V) \times P(V)$

$= 0.1 \times 0.1 \times 0.1$

$= 0.001$

(b)

$\therefore P(\text{At least 1 Voice call}) = 1 - P(DDD)$

$= 1 - [P(D) \times P(D) \times P(D)]$

$= 1 - [0.9 \times 0.9 \times 0.9]$

$= 0.271$

(c)

$\therefore P(\text{At best one voice call}) = P(VDD, DVD, DDV, DDD)$

$= P(V) \times P(V) \times P(D) + P(D) \times P(V) \times P(D) + P(D) \times P(D) \times P(V) + P(D) \times P(D) \times P(D)$

$= 0.1 \times 0.9 \times 0.9 + 0.9 \times 0.1 \times 0.9 + 0.9 \times 0.9 \times 0.1 + 0.9 \times 0.9 \times 0.9$

$= 0.972$

Exercise:-

3.16 In a Communication system, Signal are sent, where Some of the signals are faded (F) & some of reached to the destination properly (P). If power of the signals is not appropriate, then 50% signal are faded. An inspection team observed 3 Consecutive signals. Find.

(a) P(At least one signal is faded)

(b) P(At best one signal is faded)

(a)

The Sample Space

$$S = \begin{Bmatrix} FFF & FFP & FPF & FPP \\ PFF & PFP & PPF & PPP \end{Bmatrix}$$

∴ We know:-

P(At least one signal faded) $= 1 - P(PPP)$

$= 1 - \frac{1}{8}$

$= \frac{7}{8}$

(b)

P(At best one signal is faded) $= P(FPP, PFP, PPF, PPP)$

$= \frac{4}{8}$

Ans

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3.17 Find the probabilities specified in problem 3.16 if 10% signals are faded.

(a)

P(At least one signal is faded) $= 1 - P(PPP)$

$= 1 - P(P)\cdot P(P)\cdot P(P)$

$= 1 - \{0.9 \times 0.9 \times 0.9\}$

$= 0.271$

is given:–
$P(F) = 10\% = 0.1$
$\therefore P(P) = 90\% = 0.9$

(b)

P(At best one signal is faded) = P(FPP, PFP, PPF, PPP)

$= P(F) \times P(P) \times P(P) + P(P) \times P(F) \times P(P) + P(P) \times P(P) \times P(F) + P(P) \times P(P) \times P(P)$

$\therefore = 0.1 \times 0.9 \times 0.9 + 0.9 \times 0.1 \times 0.9 + 0.9 \times 0.9 \times 0.1 + 0.271$

$= 0.081 + 0.081 + 0.081 + 0.271$

$= 0.514$

problems 3.15:–

⇒ In a box a ball are present as 1, 2, 3, 4. If two balls are drawn one by one with replacement. Find

(a) P(Sum of the numbers is 5 or 1st ball drawn has number 3.)

(b) P(Sum of the num is 5 given that the second drawn ball bears the number 3).

⇒ 1st Determine the sample space:-

	1	2	3	4
1	(1,1)	(2,1)	(3,1)	(4,1)
2	(1,2)	(2,2)	(3,2)	(4,2)
3	(1,3)	(2,3)	(3,3)	(4,3)
4	(1,4)	(2,4)	(3,4)	(4,4)

(a)

Let,

Sum of the nums of two balls are 5 = Event A

1st ball drawn num as 3 = Event B

$\therefore A = \{\{4,1\}, \{3,2\}, \{2,3\}, \{1,4\}\}$

$\therefore B = \{\{3,1\}, \{3,2\}, \{3,3\}, \{3,4\}\}$

$\therefore A \cap B = \{\{3,2\}\}$

we know:-

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{4}{16} + \frac{4}{16} - \frac{1}{16}$$

$$= \frac{7}{16}$$

(b)

From question (a):-

$P(\text{Sum of the nums of two balls are } 5) = P(A) = \frac{4}{16}$

Let, 2nd ball drawn num as 3 = Event C

$\therefore C = \{\{1,3\}, \{2,3\}, \{3,3\}, \{4,3\}\}$

$\therefore$ We know:-

$$P(A/C) = \frac{P(A \cap C)}{P(C)} \quad \text{——— (1)}$$

$\therefore A \cap C = \{\{2,3\}\}$

$\therefore P(A \cap C) = \frac{1}{16}$

put it into the eq (1) =)

$$\therefore P(A/C) = \frac{\frac{1}{16}}{\frac{4}{16}} = \frac{1}{4}$$

Ans:-

Examples 3.16:-

=) Find the probabilities in problem 3.15 if ~~balls~~ are drawn without replacement.

⇒ Now Determine the sample space:—

	1	2	3	4
1		(2,1)	(3,1)	(4,1)
2	(1,2)		(3,2)	(4,2)
3	(1,3)	(2,3)		(4,3)
4	(1,4)	(2,4)	(3,4)	

[Since, without replacement]

(a)

Let,

Event A = Sum of the Nums are two balls is 5.

$\therefore A = \{\{4,1\}, \{3,2\}, \{2,3\}, \{1,4\}\}$

$\therefore P(A) = \frac{4}{12}$

Event B = 1st ball drawn num as 3

$B = \{\{3,1\}, \{3,2\}, \{3,4\}\}$

$\therefore P(B) = \frac{3}{12}$

$\therefore A \cap B = \{\{3,2\}\}$

So, $P(A \cap B) = \frac{1}{12}$

We know:—

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{4}{12} + \frac{3}{12} - \frac{1}{12}$$
$$= \frac{6}{12}$$

Ans:-

(b)

From Q: (a):-

$$P(A) = \frac{4}{12}$$

Let, Even e = 2^{nd} ball drawn num as 3

$$\therefore e = \{(1,3),(2,3),(4,3)\}$$

$$\therefore P(e) = \frac{3}{12}$$

So, $A \cap e = \{(2,3)\}$

$$\therefore P(A \cap e) = \frac{1}{12}$$

We know:-

$$P(A/e) = \frac{P(A \cap e)}{P(e)} = \frac{\frac{1}{12}}{\frac{3}{12}} = \frac{1}{3}$$

Ans:-

Examples 3.12:-

⇒ There are 7 Vivo & 5 LG mobile set in a box. Two sets are drawn at random. Find.

(a) P(both are Vivo(V) sets)

(b) P(one set is Vivo(V) other is LG)

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(a)

in give:-

Vivo = 7

LG = 5

$\therefore$ total = 7 + 5 = 12

So two sets can be selected randomly in $^{12}C_2$ ways.

$\therefore P(\text{both are vivo}) = P(VV) = \dfrac{^7C_2}{^{12}C_2}$

$= \dfrac{7}{22}$

(b)

$\therefore P(\text{one set is vivo } (V) \text{ other in } LG) = P(V, LG) = \dfrac{^7C_1 \times {}^5C_1}{^{12}C_2}$

$\therefore P(LG, V) = \dfrac{7 \times 5}{66}$

$= \dfrac{35}{66}$

Ans:-

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3.13

⇒ Repeat Example 3.12 for drawn at one by one with replacement.

(a)

In given:-

$$\text{Vivo} = 7$$

$$LG = 5$$

$$\text{total} = 7 + 5 = 12$$

Now, Determine the sample space.

$$S = \{VV, VL, LV, LL\}$$

$$\therefore P(\text{both sets are vivo}) = P(VV)$$

$$= P(V) \times P(V)$$

$$= 7/12 \times \frac{7}{12}$$

$$= \frac{49}{144}$$

Ans:-

(b)

$$\therefore P(1 \text{ is Vivo, other is } LG) = P(VL) + P(LV)$$

$$= P(V) \times P(L) + P(L) \times P(V)$$

$$= \frac{7}{12} \times \frac{5}{12} + \frac{7}{12} \times \frac{5}{12}$$

$$= \frac{35}{72}$$

Ans:-

GOOD LUCK

Example:- 3.14 :-

⇒ Find the probabilities specified in Example 3.12 for drawn at one by one without replacement.

(a)

$$P(\text{both sets are vivo}) = P(V) \times P(V)$$
$$= \frac{7}{12} \times \frac{6}{11}$$
$$= \frac{7}{22}$$

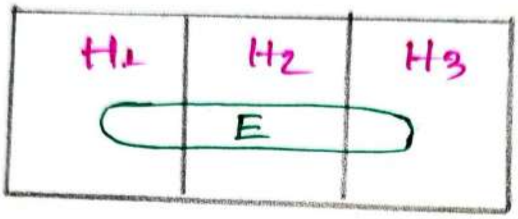
(b)

$$P(\text{one set is vivo other is LG}) = P(VL) + P(LV)$$
$$= P(V) \times P(L) + P(L) \times P(V)$$
$$= \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11}$$
$$= \frac{35}{66}$$

Ans:-

Bayes' theorem

$$\boxed{P(H_i/E) = \frac{P(E/H_i)\cdot P(H_i)}{\sum P(E/H_i)\cdot P(H_i)}}$$



$$\therefore P(H_1/E) = \frac{P(E/H_1)\cdot P(H_1)}{P(E/H_1)\cdot P(H_1) + P(E/H_2)\cdot P(H_2) + P(E/H_3)\cdot P(H_3)}$$

$$\therefore P(H_2/E) = \frac{P(E/H_2)\cdot P(H_2)}{P(E/H_1)\cdot P(H_1) + P(E/H_2)\cdot P(H_2) + P(E/H_3)\cdot P(H_3)}$$

$$\therefore P(H_3/E) = \frac{P(E/H_3)\cdot P(H_3)}{P(E/H_1)\cdot P(H_1) + P(E/H_2)\cdot P(H_2) + P(E/H_3)\cdot P(H_3)}$$

P (H_i/E) – This is the posterior probability. Posteriori basically means deriving theory out of given evidence. It denotes the conditional probability of H (hypothesis), given the evidence E.

P (E/H_i) – It is the conditional probability of the occurrence of the evidence, given the hypothesis.

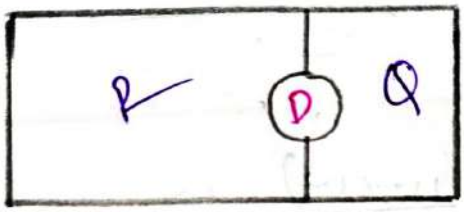
P (H) – It is the prior probability which is without the involvement of the data or the evidence.

P (E) – This is the probability of the occurrence of evidence regardless of the hypothesis.

Examples :-

⇒ Factory produces R & Q products in a ratio of 70 : 30 . 5% of R & 8% of Q are defective. From Inspection of one product its found out as depective what is the probability that is R.

⇒



To given :-

$P(R) = 70\% = 0.7$
$P(Q) = 30\% = 0.3$
$P(D/R) = 5\% = 0.05$
$P(D/Q) = 8\% = 0.08$

We know :-

$$P(R/D) = \frac{P(D/R) \times P(R)}{P(D/R) \times P(R) + P(D/Q) \times P(Q)}$$

$$= \frac{0.05 \times 0.7}{0.05 \times 0.7 + 0.08 \times 0.3}$$

$$= \frac{35}{59}$$

Problem 3.21: In a box there are 70% mathematics books and 30% electrical engineering books. Among mathematics books 40% are foreign books and among electrical engineering books 50% are foreign books. A foreign book is selected. What is the probability that the selected one is an electrical engineering book?

⇒ Let:-

Math Book = m

Electrical Eng Book = E

Foreign Book = F

In given:-

$$P(m) = 70\% = 0.7$$

$$P(E) = 30\% = 0.3$$

$$P(F/E) = 50\% = 0.5$$

$$P(F/m) = 40\% = 0.4$$

We know:-

$$P(E/F) = \frac{P(F/E) \times P(E)}{P(F/E) \times P(E) + P(F/m) \times P(m)}$$

$$= \frac{0.5 \times 0.3}{0.5 \times 0.3 + 0.4 \times 0.7}$$

$$= \frac{15}{43}$$

Ans:-

Exercise

3.1. Tickets are numbered as 1 to 20, mixed up and then one is drawn randomly. Find the probability that the ticket drawn has a number which is a multiple of 3 or 5.

In given:—

$S = \{1, 2, 3, 4, \ldots, 19, 20\}$

⇒ Multiple of 3:—

$A = \{3, 6, 9, 12, 15, 18\}$ $\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{20}$

Multiple of 5:—

$B = \{5, 10, 15, 20\}$ $\therefore P(B) = \frac{n(B)}{n(S)} = \frac{4}{20}$

$\therefore A \cap B = \{15\}$ $\therefore P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{20}$

$\therefore$ we know:—

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{6}{20} + \frac{4}{20} - \frac{1}{20}$$
$$= \frac{9}{20}$$

Ans:—

3.2. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

⇒ In given,

red = 2
green = 3
blue = 2
total = 2+3+2 = 7

∴ Number of ways of drawing 2 balls out of 7

$= {}^7C_2 = 21$

Number of balls without blue = 5

So, Number of ways of drawing 2 balls out of 5

$= {}^5C_2 = 10$

∴ P (none of the balls drawn is blue) $= \frac{{}^5C_2}{{}^7C_2} = \frac{10}{21}$

Ans:-

3.3. There are 8 red, 7 blue and 6 green balls in a box. One ball is picked up randomly. What is the probability that it is neither red nor green?

⇒ In given:-

red = 8

blue = 7

green = 6

∴ total = 8 + 7 + 6 = 21

So, if the ball drawn is neither red nor green that means. the ball drawn is blue

∴ P (drawn a blue ball) = $P(b) = \frac{7}{21}$

Ans:-

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3.4. What is the probability of getting a sum 9 from two throws of a die?

$\Rightarrow$ 1st Determine the sample space (S):-

	1	2	3	4	5	6
1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
6	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)

$\therefore$ Let

Event A = Sum of 9 from two throws of a die

$\therefore A = \{(3,6), (4,5), (5,4), (6,3)\}$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{4}{36}$

Ans.

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3.5. Three unbiased coins are tossed. What is the probability of getting at most two heads?

⇒ Is given:-

The sample space $S = \{HHH \quad HHT \quad HTH \quad HTT \quad THH \quad THT \quad TTH \quad TTT\}$

Let,

Event A = At most two head

$\therefore A = \{HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$\therefore n(A) = 7$

$n(S) = 8$

$\therefore P(\text{At most two head}) = \frac{n(A)}{n(S)} = \frac{7}{8}$

Ans:-

3.6. Two die are thrown at once. Find the probability of getting two numbers whose product is even.

⇒ From Q: 3.5:-

Let,

Event A = two number whos product is even.

$\therefore A = \{(2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (6,1), (6,2), (6,3), (6,4), (6,6), (1,2), (3,2), (5,2), (1,4), (3,4), (5,4), (1,6), (3,6), (5,6), (6,5)\}$

$\therefore n(A) = 27$

$n(S) = 36$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{27}{36} = \frac{3}{4}$

Ans:-

3.7. There are 15 boys and 10 girls in a class. Three students are selected at random. Find the probability that 1 girl and 2 boys are selected.

⇒ In given:-

Boys = 15

Girls = 10

$\therefore$ Total = 15 + 10 = 25

$\therefore n(S) = {}^{25}C_3$

$\therefore P(1 \text{ girl} \& 2 \text{ Boys}) = \frac{{}^{15}C_2 \times {}^{10}C_1}{{}^{25}C_3}$

$= \frac{21}{46}$

3.8. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?

⇒ In given:-

prizzes = 10

Blank = 25

$\therefore$ total = 35

$\therefore P(\text{getting a prizes}) = \frac{n(A)}{n(S)} = \frac{10}{35} = \frac{2}{7}$

3.9. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. Find the probability that all of them are red.

⇒ In given:-

white = 4
red = 5
blue = 6

$\therefore$ total = 15

$\therefore n(S) = {}^{15}C_3$

$$\therefore P(\text{All of them are red}) = \frac{n(\text{red})}{n(S)} = \frac{{}^{5}C_3}{{}^{15}C_3} = \frac{2}{91}$$

Ans:-

3.10. In an office there are 5 computers identified by serial number 1, 2, 3, 4 and 5. Two computers are selected by two persons who work at (i) different working hours [with replacement], (ii) same working hour [without replacement]. Find the probability that sum of the numbers of the selected computers is (a) 8 or first selected computer has the number 3, and (b) 6 given that second selected computer has the number 4.

⇒ 1st Determine the Sample space:-

	1	2	3	4	5	
1	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	→ for sum 6 → 1st selected computer has 3
2	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	
3	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	→ for sum 8
4	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	→ 2nd selected computer has 4
5	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	

(a)

(i) For with replacement:-

Let,

Event A = Sum of the nums is 8

$\therefore A = \{(3,5), (4,4), (5,3)\}$

$\therefore n(A) = 3$

$n(S) = 25$

Event B = 1st selected computer is 3.

$\therefore B = \{(3,1), (3,2), (3,3), (3,4), (3,5)\}$

$\therefore n(B) = 6 - 1 = 5$

$\therefore A \cap B = \{(3,5)\}$

$\therefore n(A \cap B) = 1$

We know:-

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)}$$

$$= \frac{3}{25} + \frac{5}{25} - \frac{1}{25}$$

$$= \frac{7}{25}$$

Ans:-

(b)

Let, Event C = Sum of the nums is 6

$\therefore C = \{(1,5), (2,4), (3,3), (4,2), (5,1)\}$

$\therefore n(C) = 5$

$n(S) = 25$

Event D = 2^{nd} selected one has 4

$\therefore D = \{(1,4), (2,4), (3,4), (4,4), (5,4)\}$

$n(D) = 5$

So, $C \cap D = \{(2,4)\}$

$\therefore n(C \cap D) = 1$

We know:—

$$P(C/D) = \frac{P(C \cap D)}{P(D)}$$

$$= \frac{\frac{1}{25}}{\frac{5}{25}}$$

$$= \frac{1}{5}$$

Ans:—

(ii) For without replacement

1st determine the Sample :-

	1	2	3	4	5
1		(2,1)	(3,1)	(4,1)	(5,1)
2	(1,2)		(3,2)	(4,2)	(5,2)
3	(1,3)	(2,3)		(4,3)	(5,3)
4	(1,4)	(2,4)	(3,4)		(5,4)
5	(1,5)	(2,5)	(3,5)	(4,5)	

→ Sum of 6

→ sum of 8

→ 2nd selected one in 4

→ 1st selected one in 3

(a)

Let,

Event A = Sum of the nums in 8

$\therefore A = \{(3,5), (5,3)\}$

$\therefore n(A) = 2$

$n(S) = 20$

Event B = 1st one in 3

$\therefore B = \{(3,1), (3,2), (3,4), (3,5)\}$

$\therefore n(B) = 4$

$\therefore A \cap B = \{(3,5)\}$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{2}{20} + \frac{4}{20} - \frac{1}{20} = \frac{5}{20} = \frac{1}{4}$

6

Let, Event C = Sum of the nums is 6

$\therefore C = \{(1,5), (2,4), (4,2), (5,1)\}$

$\therefore n(C) = 4$

$n(S) = 20$

Event D = 2^{nd} selected one. is 4

$\therefore D = \{(1,4), (2,4), (3,4), (5,4)\}$

$\therefore n(D) = 4$

Sol

$C \cap D = \{(2,4)\}$

$\therefore n(C \cap D) = 1$

We Know:-

$$P(C/D) = \frac{P(C \cap D)}{P(D)}$$

$$= \frac{\frac{1}{20}}{\frac{4}{20}}$$

$$= \frac{1}{4}$$

3.11. There are 5 electronic engineers and 6 computers engineers in a mobile operator's office. A committee of 4 is to be formed to perform a duty. Find the probability that the committee will consist of (a) all electronic engineers, and (b) 2 electronic engineers and 2 comput[illegible] [illegible]ineers.

⇒ In given

Electronic Eng (E) = 5

Computers Eng (C) = 6

$\therefore$ total = 5+6 = 11

$\therefore n(S) = {}^{11}C_4$

(a)

$$P(\text{All } E) = \frac{{}^5C_4}{{}^{11}C_4} = \frac{1}{66}$$

(b)

$$P(2E \,\&\, 2C) = \frac{{}^5C_2 \times {}^6C_2}{{}^{11}C_4}$$

$$= \frac{5}{11}$$

Ans:-

3.12. In a packet there are 7 Samsung and 5 Nokia mobile phone sets. Two sets are drawn one after another with replacement. Find the probability that (a) both are Samsung sets, and (b) one set is Samsung and another one is Nokia.

=> In given:-

Samsung (S) = 7

Nokia (N) = 5

$\therefore$ total = 7+5 = 12

(a)

$$P(\text{both sets are Samsung}) = P(SS)$$
$$= P(S) \times P(S)$$
$$= \frac{7}{12} \times \frac{7}{12}$$
$$= \frac{49}{144}$$

(b)

P(1 set are Samsung and Another one is Nokia)

$$= P(SN) + P(NS)$$
$$= P(S) \times P(N) + P(N) \times P(S)$$
$$= \frac{7}{12} \times \frac{5}{12} + \frac{5}{12} \times \frac{7}{12}$$
$$= \frac{35}{72}$$

3.13. Find the probabilities specified in problem 3.12 if sets are drawn without replacement.

(a)

$$P(\text{both are Samsung set}) = P(SS) = P(S) \times P(S) = \frac{7}{12} \times \frac{6}{11} = \frac{7}{22}$$

(b)

$$P(\text{1 set in Samsung Another 1 is Nokia}) = P(SN) + P(NS) = P(S) \times P(N) + P(S) \times P(N) = \frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{35}{66}$$

3.14. 75% signals sent from a server reach to its goal properly. Once 3 signals are checked randomly. Find the probability that out of 3, (a) at least 1 reaches properly, and (b) at best 2 reach properly.

Let, Reach properly = A

Not Reach " = B

In Given:-

$$P(A) = 75\% = 0.75$$
$$P(B) = 25\% = 0.25$$

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1st determine the sample space:-

$$S = \left\{ \begin{matrix} AAA, & AAB, & ABA, & ABB \\ BAA, & BAB, & BBA, & BBB \end{matrix} \right\}$$

(a)

P(At least one Reach properly) $= 1 - P(BBB)$

$= 1 - P(B) \times P(B) \times P(B)$

$= 1 - 0.25 \times 0.25 \times 0.25$

$= 0.9843$

(b)

P(At best two reach properly) $= P(AAB, ABA, ABB, BAA, BAB, BBA, BBB) = 1 - P(AAA)$

$= 1 - \{P(A) \times P(A) \times P(A)\}$

$= 1 - \{0.75 \times 0.75 \times 0.75\}$

$= 0.5781$

Ans:-

3.15. There are 50 computers in an office. Of them, 20 are ACER and 30 are Dell. The computers are investigated and found that 12 ACER and 25 Dell computers are good. One computer is selected at random. Find the probability that the selected computer is (a) either Dell or good, and (b) ACER given that it is good.

⇒ Is given mon

Dell = 30

ACER = 20

∴ total = 30 + 20 = 50

Good ACER (GA) = 12

∴ Bad " (BA) = 8

Good Dell (GD) = 25

∴ Bad Dell (BD) = 5

let,

$P(Dell) = P(D) = \frac{30}{50}$

$P(ACER) = P(A) = \frac{20}{50}$

$\therefore P(Good) = P(G) = \frac{37}{50}$

total Good = 25 + 12 = 37

$\therefore P(D \cap G) = \frac{25}{50}$

$P(A \cap G) = \frac{12}{50}$

(a)

$$\therefore P(D \cup G) = P(D) + P(G) - P(D \cap G)$$

$$= \frac{30}{50} + \frac{37}{50} - \frac{25}{50}$$

$$= \frac{21}{25}$$

(b)

$$P(A/G) = \frac{P(A \cap G)}{P(G)} = \frac{\frac{12}{50}}{\frac{37}{50}} = \frac{12}{37}$$

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Problem 3.17: Suppose there are 30 students, out of which 12 are from EEE department and 18 are from CSE department. The CGPA of 8 EEE and 12 CSE students are found to be good. One student is selected randomly. Find the probability that the selected student is (a) from EEE dept. given that his CGPA is not good, and (b) from CSE dept. or his CGPA is good.

⇒ In given:—

EEE(E) = 12

CSE(C) = 18

Total = 12+18 = 30

Good CGPA of EEE (GE) = 8

Good CGPA of CSE (GC) = 12

∴ total Good CGPA (TGC) = 8+12 = 20

Not Good CGPA of EEE (NGE) = 12−8 = 4

Not " " " CSE (NGC) = 18−12 = 8

∴ total Not Good CGPA (TNGC) = 4+8 = 10

Now,

$P(E) = 12/30$

$P(C) = 18/30$

$P(TGC) = \frac{20}{30}$

$P(TNGC) = \frac{10}{30}$

$\therefore P(E \cap NGE) = \frac{4}{30}$

$\therefore P(C \cap GC) = \frac{12}{30}$

(a)

$$P(E/TNGC) = \frac{P(E \cap NGE)}{P(TNGE)} = \frac{\frac{4}{30}}{\frac{10}{30}} = \frac{4}{10}$$

(b)

$$P(C \cup TGC) = P(C) + P(TGC) - P(C \cap GC)$$

$$= \frac{18}{30} + \frac{20}{30} - \frac{12}{30} = \frac{26}{30}$$

3.19. While watching a game of Champions League football in a cafe, you observe someone who is clearly supporting Manchester United in the game. What is the probability that they were actually born within 25 miles of Manchester? Assume that the probability that a randomly selected person in a typical local bar environment is born within 25 miles of Manchester is 1/20, the chance that a person born within 25 miles of Manchester actually supports United is 7/10, and the probability that a person not born within 25 miles of Manchester supports United with probability 1/10.

⇒ In given

$$P(B) = \frac{1}{20}$$

$$P(\text{Not Born}) = P(N) = 1 - \frac{1}{20} = \frac{19}{20}$$

$$P(S/B) = \frac{7}{10}$$

$$P(S/N) = \frac{1}{10}$$

We know—

$$P(B/S) = \frac{P(S/B) \times P(B)}{P(S/B) \times P(B) + P(S/N)\,P(N)}$$

$$= \frac{\frac{7}{10} \times \frac{1}{20}}{\frac{7}{10} \times \frac{1}{20} + \frac{1}{10} \times \frac{19}{20}}$$

$$= \frac{7}{26}$$

Ans

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#Random Variable:

Ch#04

"GOOD LUCK"

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Random Varriable:—

⇒ A rrandom varriable is a varriable whose possible values are numerrical outcomes of a rrandom experiment. Usally written as X.

There are two types:—

1. Discrete random varriable.

2. Continuous random varriable.

Discreate Random Varriable

Formula:—

1. $\sum_{x=0}^{\infty} P(x) = 1$

2. $0 \leq P(x) \leq 1$

Continuous Random Varriables—

Formula:—

1. $\int_{-\infty}^{\infty} f(x)\,dx = 1$

2. $f(x) \geq 0$

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Joint probability function:-

For Discrete:-

1. $p(x,y) \geq 0$
2. $\sum_{x=0}^{\infty} \sum_{y=0}^{\infty} p(x,y) = 1$

For Continuous:-

1. $f(x,y) \geq 0$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$

Formula:-

⇒ **Mean: (Expected values):- E(x)**

1. $E(x) = \sum x \cdot p(x) \rightarrow$ if x is descrete
2. $E(x) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx \rightarrow$ if x is continuos.

⇒ **Variances:-**

1. $V(x) = \sigma^2 = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$

$\therefore$ 1.1 $V(x) = E(x^2) - [E(x)]^2$

Where,

$E(x^2) = \sum x^2 \cdot p(x) \rightarrow$ for desereate

$E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) \cdot dx \rightarrow$ for Continuous.

#properties for Random Variable:-

Constant

1. $E(A) = A$
2. $V(A) = 0$
3. $E(AX) = A\,E(X)$
4. $V(AX) = A^2 V(X)$
5. $E(AX \pm BY) = A\,E(X) \pm B\,E(Y)$
6. $V(AX \pm BY) = A^2 V(X) + B^2 V(Y) \pm 2AB\,\mathrm{COV}(XY)$
7. $\mathrm{COV}(x,y) = E(XY) - E(X)\cdot E(Y)$

Note:- if x, y are independent

Then:- 7.1:- $\mathrm{COV}(x,y) = 0$

$\therefore$ 6.1:- $V(AX \pm BY) = A^2 V(X) + B^2 V(Y)$

8. $h(y)\cdot g(x) = f(x,y)$

so, $h(y)$ & $g(x)$ are independent.

Example: 4.1:-

X	0	1	2	3	4	5	total
P(X)	0.28	0.37	0.23	0.08	0.02	0.02	1

⇒ Find the probability that

(a) More than Once.

(b) Less or two times.

(c) Four or more times.

⇒

(a)

$$P(x>1) = P(x=2) + P(x=3) + P(x=4) + P(x=5)$$

$$= 0.23 + 0.09 + 0.02 + 0.01$$

$$= 0.35$$

(b)

$$P(x\le 2) = P(x=0) + P(x=1) + P(x=2)$$

$$= 0.28 + 0.37 + 0.23$$

$$= 0.88$$

(c)

$$P(x\ge 4) = P(x=4) + P(x=5)$$

$$= 0.02 + 0.01$$

$$= 0.03$$

(d) $E(2X+1)$

(e) $V(4x-3)$

(f) $V(x)$

⇒

(d)

$$E(2X+1) = 2E(X) + E(1)$$
$$= 2 \times \sum x \cdot p(x) + 1$$
$$= 2\{[0\times 0.28 + 1\times 0.37 + 2\times 0.23 + 3\times 0.09 + 4\times 0.02 + 5\times 0.01]\} + 1$$
$$= 2\times\{0.37 + 0.46 + 0.27 + 0.08 + 0.05\} + 1$$
$$= 3.46$$

(e)

$$V(4x-3) = 4^2 V(x) - V(3)$$
$$= 16\{E(x^2) - [E(x)]^2\} \quad \text{—— (1)} \qquad [\because V(3) = 0]$$

we have
now
$$E(x) = 1.23$$
$$\therefore E(x^2) = \sum x^2 \cdot p(x)$$
$$= (0^2\times 0.28 + 1^2\times 0.37 + 2^2\times 0.23 + 3^2\times 0.09 + 4^2\times 0.02 + 5^2\times 0.01)$$
$$= 0.37 + 0.92 + 0.81 + 0.32 + 0.25$$
$$= 2.67$$

Now put the values in eq ① =>

$$V(x \times 4 - 3) = 4^2 V(x) - 0$$
$$= 16 \times \{E(x^2) - (E(x))^2\}$$
$$= 16 \times \{2.62 - 1.23^2\}$$
$$= 17.71$$

Ans:-

(E)

$$V(x) = E(x^2) - \{E(x)\}^2$$
$$= 2.62 - (1.23)^2$$
$$= 1.1071$$

Ans:-

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Examples 4.10:—

y \ x	1	2	3	4	5	6	g(x)
1	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{3}$
2	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{3}$
3	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{3}$
h(y)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1

(a) Marginal distribution of (X)

(b) Marginal distribution of (Y)

(c) $E(x)$

(d) $E(y)$

(a)

⇒ Marginal distribution of X:—

X	1	2	3	total
g(x)	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

(b)

⇒ Marginal distribution of Y:—

Y	1	2	3	4	5	6	total
h(Y)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1

(c)

We know:—

$$E(x) = \sum x \cdot g(x)$$

$$= \left[1\times\frac{1}{3} + 2\times\frac{1}{3} + 3\times\frac{1}{3}\right]$$

$$= 2$$

(d)

We know:—

$$E(Y) = \sum y \cdot h(y)$$

$$= \left[\frac{1}{6}\times 1 + 2\times\frac{1}{6} + 3\times\frac{1}{6} + 4\times\frac{1}{6} + 5\times\frac{1}{6} + \frac{1}{6}\times 6\right]$$

$$= \frac{21}{6} = \frac{7}{2} = 3.5$$

(e) $E(5X-Y)$

(f) $V(Y-2X)$

(e)

$$E(5x-Y) = 5E(x) - E(Y)$$

$$= 5\times[2] - 3.5$$

$$= 6.5$$

(f)

$$V(Y-2X) = V(Y) - 2^2 V(x)$$

$$= \{E(Y^2) - (E(Y))^2\} - 2^2\{E(x^2) - (E(x))^2\}$$

(1)

$$\therefore E(Y^2) = \sum Y^2 \cdot h(y)$$

$$= \left\{ 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + 3^2 \times \frac{1}{6} + 4^2 \times \frac{1}{6} + 5^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} \right\}$$

$$= 15.167$$

$$E(x^2) = \sum x^2 \cdot g(x)$$

$$= \left\{ 1^2 \times \frac{1}{3} + 2^2 \times \frac{1}{3} + 3^2 \times \frac{1}{3} \right\}$$

$$= 4.667$$

put the values in eq (1) 2)

$$V(Y-2X) = \left\{ E(Y^2) - (E(Y))^2 \right\} - 2^2 \cdot \left\{ E(x^2) - (E(x))^2 \right\}$$

$$= \left\{ 15.167 - (3.5)^2 \right\} - 2^2 \times \left\{ (4.667 - 2^2) \right\}$$

$$= 2.917 - 2.668$$

$$= 0.249$$

Ans:-

Examples 4.6:

$$f(x) = \frac{x}{2} \; ; \; 0 < x < 2$$

Find

(I) $P(x < 1)$

(II) $P(x > 0.5)$

(III) $P(x \geq 1)$

(IV) $P(1 < x < 2)$

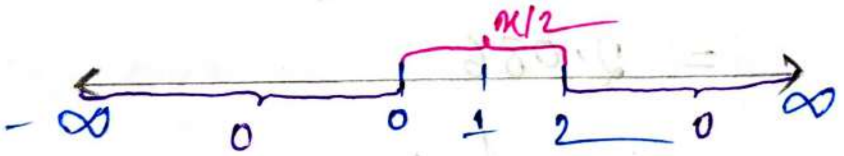
(V) $E(x)$

(VI) $V(x)$

(a)

Is given:

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$



$$P(x < 1) = \int_0^1 f(x) \cdot dx$$

$$= \int_0^1 \left(\frac{x}{2}\right) \cdot dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{4}$$

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(b)

$$P(x > 0.5) = \int_{0.5}^{2} f(x) \cdot dx$$

$$= \int_{0.5}^{2} \left[\frac{x}{2}\right] \cdot dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_{0.5}^{2}$$

$$= \frac{1}{2} \times \left\{2 - 0.125\right\}$$

$$= 0.9375$$

(c)

$$P(x > 1) = \int_{1}^{2} f(x) \cdot dx$$

$$= \int_{1}^{2} \left[\frac{x}{2}\right] dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_{1}^{2}$$

$$= \frac{1}{2} \times \left[2 - \frac{1}{2}\right]$$

$$= 0.75$$

(d)

$$P(1 < x < 2) = \int_{1}^{2} f(x) \cdot dx$$

$$= 0.75$$

(e)

we know:—

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx$$

$$= \int_0^2 \left(x \times \frac{x}{2}\right) \cdot dx$$

$$= \frac{1}{2} \int_0^2 \left(\frac{x^2}{1}\right) \cdot dx$$

$$= \frac{1}{2} \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2} \times \frac{8}{3}$$

$$= \frac{4}{3}$$ Ans:-

(f)

we know:—

$$V(x) = E(x^2) - (E(x))^2 \quad \text{--- (1)}$$

$$\therefore E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) \cdot dx$$

$$= \int_0^2 \left(x^2 \times \frac{x}{2}\right) \cdot dx$$

$$= \int_0^2 \frac{x^3}{2} dx$$

$$= \frac{1}{2} \times \left[\frac{x^4}{4}\right]_0^2$$

$$= \frac{1}{2} \times \frac{2^4}{4} = 2$$

∴ put the values in eq ①

$$\therefore V(x) = E(x^2) - (E(x))^2$$

$$= 2 - \left(\frac{4}{3}\right)^2$$

$$= \frac{2}{9}$$

Ans:-

problems 4.7:-

Given that the join probability density

$$f(x,y) = 4x(1-y), \quad 0<x<1;\ 0<y<1$$

ⓐ Marginal distribution of X

ⓑ marginal distribution of Y.

ⓒ Show that X & Y are independent.

ⓐ

$$\boxed{g(x) = \int_y f(x,y) \cdot dy}$$

$$= \int_0^1 4x(1-y) \cdot dy$$

$$= 4x \int_0^1 (1-y) \cdot dy$$

$$= 4x \left[y - \frac{y^2}{2}\right]_0^1$$

$$g(x) = 4x - 2x = 2x$$

(b)

$$\boxed{h(y) = \int_x f(x,y) \cdot dx}$$

$$= \int_0^1 4x(1-y) \cdot dx$$

$$= (1-y) \times 4 \int_0^1 x \cdot dx$$

$$= 4(1-y) \left[\frac{x^2}{2}\right]_0^1$$

$$\therefore h(y) = 2(1-y)$$

(c)

We know,

if $g(x) \times h(y) = f(x,y)$

Then $g(x)$ & $h(y)$ independent

$\therefore$ L.H.S:-

$$= g(x) \times h(y)$$

$$= 2x \times 2(1-y)$$

$$= 4x(1-y)$$

$$= f(x,y)$$

$$= R.H.S$$

So, $g(x)$ & $h(y)$ are Independent.

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(d) Find $V(2x-y)$

(d)

$$V(2x-y) = 2^2 V(x) - V(y)$$
$$= \{4[E(x^2)] - \{E(x)\}^2\} - \{E(y^2) - (E(y))^2\} \quad (1)$$

Now, we know:—

$$E(x) = \int_0^1 x \cdot g(x) \cdot dx$$
$$= \int_0^1 x \cdot 2x \cdot dx$$
$$= 2\int_0^1 x^2 \cdot dx$$
$$= 2\left[\frac{x^3}{3}\right]_0^1 dx$$
$$= \frac{2}{3}$$

$$E(x^2) = \int_0^1 x^2 \cdot g(x) \cdot dx$$
$$= \int_0^1 x^2 \cdot 2x \cdot dx$$
$$= \int_0^1 2x^3 \cdot dx$$
$$= 2\left[\frac{x^4}{4}\right]_0^1$$
$$= \frac{1}{2}$$

$$E(y) = \int_0^1 y \cdot h(y) \cdot dy$$
$$= \int_0^1 y \cdot 2(1-y) \cdot dy$$
$$= \int_0^1 (2y - 2y^2) \cdot dy$$
$$= 2\times\left[\frac{y^2}{2}\right]_0^1 - 2\times\left[\frac{y^3}{3}\right]_0^1$$
$$= 1 - \frac{2}{3}$$
$$= \frac{1}{3}$$

$$E(y^2) = \int_0^1 y^2 \cdot h(y) \cdot dy$$
$$= \int_0^1 y^2 \cdot 2(1-y) \cdot dy$$
$$= \int_0^1 (2y^2 - 2y^3)\, dy$$
$$= 2\left[\frac{y^3}{3}\right]_0^1 - 2\left[\frac{y^4}{4}\right]_0^1$$
$$= \frac{2}{3} - \frac{1}{2}$$
$$= \frac{1}{6}$$

Now put All values in eq ① ⇒

$$V(2x-y) = \left\{4 \times \frac{1}{2} - \left(\frac{2}{3}\right)^2\right\} - \left\{\frac{1}{6} - \left(\frac{1}{3}\right)^2\right\}$$

$$= \frac{14}{9} - \frac{1}{18}$$

$$= \frac{3}{2}$$

Ans:-

Examples 4.5:-

$$f(x) = \begin{cases} k(x^2+x) & ; 0 < x < 1 \\ 0 & ; \text{otherwise} \end{cases}$$

(a) Find value of k

(b) Find average time

(a)

we know:-

$$\Rightarrow \int_0^1 f(x) \cdot dx = 1$$

$$\Rightarrow \int_0^1 k(x^2+x) \cdot dx = 1$$

$$\Rightarrow k\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_0^1 = 1$$

$$\Rightarrow k\left[\frac{1}{3} + \frac{1}{2}\right] = 1$$

$\therefore K = \frac{6}{5}$

(b)

We know:—

$$E(x) = \int_0^1 x \cdot f(x) \cdot dx$$

$$= \int_0^1 x \cdot K(x^2 + x) \cdot dx$$

$$= \frac{6}{5} \int_0^1 (x^3 + x^2) \cdot dx$$

$$= \frac{6}{5} \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_0^1$$

$$= \frac{7}{10}$$

Ans:-

Examples 4.6:—

$$f(x) = \frac{x}{2} \; ; \quad 0 < x < 2$$

Find:-

(i) $P(x < 1)$

(ii) $P(x > 0.5)$

(iii) $P(1 < x < 2)$

(i)

$$P(x<1) = \int_0^1 f(x)\cdot dx$$

$$= \int_0^1 \frac{x}{2}\cdot dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{4}$$

Ans:-

(ii)

$$P(x>0.5) = \int_{0.5}^2 f(x)\cdot dx$$

$$= \int_{0.5}^2 \frac{x}{2}\cdot dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_{0.5}^2$$

$$= \frac{1}{2}\left[2 - \frac{1}{8}\right]$$

$$= \frac{15}{16}$$

$$= 0.9375$$

Ans:-

(iii)

$$P(1<x<2) = \int_1^2 f(x)\cdot dx$$

$$\therefore \quad = \int_1^2 \frac{x}{2}\cdot dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_1^2$$

$$= \frac{1}{2}\left[\frac{4}{2} - \frac{1}{2}\right]$$

$$= \frac{3}{4}$$

Ans:-

DISCRETE DISTRIBUTION

Ch: 05

Discrete Distribution:-

Probality Distribution:-

⇒ A probabily distribution is a list of all the possible outcomes. of a random variable along with their corresponding probability values.

there are two types:-

1. Discrete probability distribution.
2. Continuous probability distribution.

Discrete probability Distribution:-

I = Identification factor

I { 1. $n < 30$, 2. two outcomes 3. P larger → **Binomial P.D**

$$P(x) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x}$$

I { 1. $n > 30$ 2. Average given 3. P small 4. outcome with a period. → **Poisson P.D**

$$P(x) = \frac{\mu^{x} \cdot e^{-\mu}}{x!}$$

1. Out come on fixed trial } I → **Geometric P.D**

$$P(x) = p(1-p)^{x-1}$$

Binomial P.D

n = Sample size
x = num. of Success.
p = probability of Success.
$E(x) = np$
$V(x) = npq$

mean > Variance

Possion P.D

x = num of outcomes
$\mu = \lambda$ = mean/Average
$E(x) = np$
$V(x) = np$

mean = Variance

Geometric P.D

p = probability of Success.
$E(x) = \frac{q}{p}$
$V(x) = \frac{q}{p^2}$

Example: 1:-

⇒ 85% knows Mac. Donald's. Randomly picked 8 students. Find the probability that 5 students knows McDonalds.

⇒ In given:-

$p = 85\% = 0.85$
$n = 8$
$x = 5$

We know:-

$$P(x=5) = {}^{n}C_{x} \times p^{x}(1-p)^{n-x}$$

$$= {}^{8}C_{5} \times (0.85)^{5} \times (1-0.85)^{8-5}$$

$$= 0.083$$

for % $= 0.083 \times 100\%$
$= 8.3\%$

Example : 1.1 :-

⇒ From Ex-1, Now Find the probability that 5 Stuenent don't know McDonalds.

In given:-

$$p = (100-85)\% = 15\% = 0.15$$

$$n = 8$$

$$x = 5$$

We know:-

$$P(x=5) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x}$$

$$= {}^{8}C_{5}\,(0.15)^{5}(1-0.15)^{8-5}$$

$$= 2.611 \times 10^{-3}$$

$$\text{For } \% = 2.611 \times 10^{-3} \times 100\%$$

$$= 0.261\%$$

ⓐ $E(x) = ?$

ⓑ $V(x) = ?$

ⓐ

We know:-

$$E(x) = n \times p = 8 \times 0.15 = 1.2$$

ⓑ

$$V(x) = npq = 8 \times 0.15 \times 0.85$$

$$= 1.02$$

Ans:-

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Example 2:-

⇒ In less than 100 years 530 Hurricane have been recorded. Find the probability that

(a) One randomly picked year faced two hurricane.

(b) $E(x)$

(c) $V(x)$

(a)

In given:-

$$\mu = \lambda = \frac{530}{100} = 5.3$$

$$n = 100$$

$$x = 2$$

We know:-

$$P(x=2) = \frac{\mu^x e^{-\mu}}{x!}$$

$$= \frac{(5.3)^2 \times e^{-5.3}}{2!}$$

$$= 0.07$$

for % $= 0.07 \times 100\% = 7\%$

(b)

$$V(x) = E(x) = np = 100 \times 0.07$$

$$= 7$$

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Examples - 3

⇒ Find the probability to have first 6 (1^{st} 6) on 4th trial of rolling a dice.

⇒ Is given:-

$$p = \frac{1}{6}$$

$$x = 4$$

We know:-

$$P(x=4) = p(1-p)^{x-1}$$

$$= \frac{1}{6}\left(1-\frac{1}{6}\right)^{4-1}$$

$$= 0.096$$

for % $= 0.096 \times 100\%$

$= 9.64\%$

(a) Find $E(x)$

(b) Find $V(x)$

Here,

(a)

$$p = \frac{1}{6}$$

$$q = 1 - \frac{1}{6} = \frac{5}{6}$$

$$\therefore E(x=4) = \frac{q}{p} = \frac{5/6}{1/6} = 5$$

(b)

$$V(x=4) = \frac{q}{p^2} = \frac{5/6}{(1/6)^2} = 30$$

Example 5.1: A fair coin is tossed 8 times. Find the probability that exactly 3 tails will befall.

⇒ In given

$n = 8$

$x = 3$

$p = q = 0.5$

we know:—

$$P(x=3) = {}^nC_x\, p^x (1-p)^{n-x}$$

$$= {}^8C_3 (0.5)^{3} \times (1-0.5)^{8-3}$$

$$= 0.219$$

Example 5.2: A student randomly guesses 5 questions; each question has 5 possible choices. Find the probability that he guesses exactly 3 correctly.

⇒ In given:—

$n = 5$

$x = 3$

$p = \frac{1}{5} = 0.2$

$q = 1 - \frac{1}{5} = 0.8$

we know:—

$$P(x=3) = {}^nC_x\, p^x \cdot (1-p)^{n-x}$$

$$= {}^5C_3 (0.2)^3 \cdot (1-0.2)^{5-3}$$

$$= 0.05$$

Example 5.3; Apollo hospital records show that 75% of patients suffering from kidney disease die of it. What is the probability that out of 6 randomly selected patients, 4 will recover?

$\Rightarrow$ In given:—

$$p = (100 - 75)\% = 0.25$$
$$n = 6$$
$$x = 4$$

we know:—

$$P(x=4) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x}$$
$$= {}^{6}C_{4}(0.25)^{4}(1-0.25)^{6-4}$$
$$= 0.033$$

Ans:—

Example 5.4: A survey found that 30% AIUB students earn money from part-time job. If 5 students are selected at random, find the probability that at least 3 of them have part-time job.

$\Rightarrow$ In given:—

$$p = 30\% = 0.3$$
$$n = 5$$
$$x = 3, 4, 5$$

$$\therefore P(x=3) = {}^{5}C_{3}(0.3)^{3} \times (1-0.3)^{5-3} = 0.1323$$
$$P(x=4) = {}^{5}C_{4}(0.3)^{4} \times (1-0.3)^{5-4} = 0.0283$$
$$P(x=5) = {}^{5}C_{5}(0.3)^{4} \times (1-0.3)^{5-5} = 2.43 \times 10^{-3}$$

$$\therefore P(x \geq 3) = P(x=3) + P(x=4) + P(x=5)$$
$$= 0.1323 + 0.0283 + 2.43 \times 10^{-3}$$
$$= 0.16303.$$

Example 5.6: Vehicles pass through Kuril junction at an average rate of 300 per hour. Find the probability that no vehicle passes in a given minute.

⇒ In given:-

$$\mu = \lambda = \frac{300}{60} = 5$$

$$x = 0$$

we know:-

$$P(x=0) = \frac{\mu^x \times e^{-\mu}}{x!} = \frac{5^0 \times e^{-5}}{0!} = e^{-5}$$

Example 5.7: RFL assembles electric motors. The probability that a motor is defective is 0.01. What is the probability that a random sample of 200 motors will contain exactly 4 defective motors?

⇒ In given:-

$$\lambda = 200 \times 0.01 = 2$$

$$x = 4$$

we know:-

$$P(x=4) = \frac{\lambda^x \times e^{-\lambda}}{x!} = \frac{2^4 \times e^{-2}}{4!} = 0.090$$

Ans:-

Example 5.8: Electricity fails according to Poisson distribution with average of 4 failures per 20 weeks in AIUB. Find the probability that there will be no electricity failure in a specific week.

⇒ Is given:-

$$\lambda = 4/20 = 0.2$$

$$x = 0$$

we know:-

$$P(x=0) = \frac{\lambda^x \times e^{-\lambda}}{x!} = \frac{(0.2)^0 \times e^{-0.2}}{0!} = e^{-0.2}$$

Example 5.9: The average number of signals sent from Kamalapur railway station, not reaching properly to Noakhali railway station, is 3 per day. Find the probability that on a particular day, the number of signals not reaching properly is (a) at best 2, and (b) at least 3.

⇒ Here, $\lambda = 3$

$x = 0, 1, 2, 3, \ldots$

(a)

$$P(x \le 2) = P(x=0) + P(x=1) + P(x=2)$$

$$= \frac{\lambda^x e^{-\lambda}}{x!} + \frac{\lambda^x e^{-\lambda}}{x!} + \frac{\lambda^x e^{-\lambda}}{x!}$$

$$= \frac{3^0 \times e^{-3}}{0!} + \frac{3^1 e^{-3}}{1!} + \frac{3^2 e^{-3}}{2!}$$

$$= e^{-3}\left[1 + \frac{3^1}{1} + \frac{3^2}{2}\right]$$

$$= 0.4232$$

(b)

$$P(x \ge 3) = 1 - P(x \le 2)$$

$$= 1 - .4232$$

$$= 0.5768$$

Ans:-

Exercise: 5:-

5.3. 80% electric circuits in AIUB work properly. One day 10 circuits were selected randomly. What was the expected number of devices working properly along its variance?

$\Rightarrow$ In given:-

$$p = 80\% = 0.8$$

$$\therefore q = (100-80)\% = 0.2$$

$$n = 10$$

We know:-

$$E(x) = np = 10 \times 0.8 = 8$$

$$V(x) = npq = 10 \times 0.8 \times 0.2 = 1.6$$

Ans:-

5.4. In a class 40% are female students. Find the probability that the 9th student enter in the class will be a male.

$\Rightarrow$ Is given:-

$$p = (100-40)\% = 60\% = 0.6$$

$$x = 9$$

We know:-

$$p(x=9) = p(1-p)^{x-1}$$

$$= 0.6(1-0.6)^{9-1}$$

$$= 3.932 \times 10^{-6}$$

Ans:

5.5. 70% of all business startups in the IT industry report that they generate a profit in their first year. If a sample of 10 new IT business startups is selected, find the probability that exactly 7 will generate a profit in their first year.

⇒ Is given:—

$$p = 70\% = 0.7$$

$$n = 10$$

$$x = 7$$

We know:—

$$P(x=7) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x}$$

$$= {}^{10}C_{7}(0.7)^{7}(1-0.7)^{10-07}$$

$$= 0.2668$$

Ans:—

5.6. Two percent handsets produced by Walton are usually found defective. Walton produces 300 sets per day. Find the probability that there will be (a) 4 defective sets (b) at best 3 defective sets (c) 2 to 5 defective sets in a day's production.

⇒ Is given:—

$$\therefore \lambda = 2\% \times 300 = 6$$

(a)

$$x = 4$$

$$\therefore P(x=4) = \frac{\lambda^{x} e^{-\lambda}}{x!} = \frac{6^{4} \times e^{-6}}{4!} = 0.1338$$

(b)

$$P(x \leq 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$= \frac{6^{0} e^{-6}}{0!} + \frac{6^{1} \times e^{-6}}{1!} + \frac{6^{2} e^{-6}}{2!} + \frac{6^{3} e^{-6}}{3!}$$

$$= e^{-6}\left(\frac{1}{1} + \frac{6^{1}}{1} + \frac{6^{2}}{2} + \frac{6^{3}}{6}\right)$$

$= e^{-6} \times 61$

$= 0.1512$

c

$P(2 \le x \le 5) = P(x=2) + P(x=3) + P(x=4) + P(x=5)$

$= \frac{6^2 \times e^{-6}}{2!} + \frac{6^3 \times e^{-6}}{3!} + \frac{6^4 \times e^{-4}}{4!} + \frac{6^5 \times e^{-5}}{5!}$

$= e^{-6}\left(\frac{6^2}{2!} + \frac{6^3}{3!} + \frac{6^4}{4!} + \frac{6^5}{5!}\right)$

$= 172.8 e^{-6}$

$= 0.4283$

Ans:-

5.7. A box of candies has many different colors in it. There is a 15% chance of getting a pink candy. What is the probability that at least 2 candies in a box are pink out of 8?

⇒ Is given:-

$p = 0.15$

$n = 8$

$\therefore P(x \ge 2) = 1 - \{P(x < 2)\}$

$= 1 - [P(x=0) + P(x=1)]$

$= 1 - [{}^{8}C_{0}(0.15)^0 \times (1-0.15)^{8-0} + {}^{n}C_{1}(0.15)^1(1-0.15)^{8-1}]$

$= 1 - [0.2724 + 0.3846]$

$= 0.343$

Ans:-

5.8. In a typical El Clasico game against arch-rival Barcelona, Real Madrid coach Zidane can expect 2 injuries on average. Find the probability that Madrid will have at most 1 injury in the game.

⇒ In given:–

$$\lambda = 2$$

$$\therefore P(x \le 1) = P(x=0) + P(x=1)$$

$$= \frac{\lambda^x e^{-\lambda}}{x!} + \frac{\lambda^x e^{-\lambda}}{x!}$$

$$= \frac{2^0 e^{-2}}{0!} + \frac{2^1 e^{-2}}{1!}$$

$$= e^{-2}(1+2)$$

$$= 3e^{-2}$$

$$= 0.4060$$

Ans:–

5.9. Let X be the number of typos on a printed page with a mean of 4 typos per page. What is the probability that a randomly selected page has a) at least two typo b) at best 3 typo c) exactly 5 typos on it?

⇒ In given:–

$$\lambda = 4$$

(a)

$$P(x \ge 2) = 1 - P(x < 2)$$

$$= 1 - [P(x=0) + P(x=1)]$$

$$= 1 - \left[\frac{e^{-4} \times 4^0}{0!} + \frac{e^{-4} \times 4^1}{1!}\right]$$

$$= 1 - e^{-4}\left(\frac{1}{0!} + \frac{4}{1!}\right)$$

$$= 0.9084$$

(b)

$$P(x \le 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$= \frac{e^{-4} \times 4^0}{0!} + \frac{e^{-4} \times 4^1}{1!} + \frac{e^{-4} \times 4^2}{2!} + \frac{e^{-4} \times 4^3}{3!}$$

$$= e^{-4}\left(1 + \frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!}\right)$$

$$= \frac{71}{3} e^{-4}$$

$$= 0.433$$

(c)

$$P(x=5) = \frac{e^{-4} \times 4^5}{5!} = 0.156$$

5.10. The number of constructions related accidents per working week follows Poisson distribution with mean 1. Find the probability that in a particular week, there will be (a) less than 2 accidents, and (b) more than 1 accidents.

⇒ In given

$\lambda = 1$

(a)

$$P(x<2) = P(x=0) + P(x=1)$$

$$= \frac{e^{-1} \times 1^0}{0!} + \frac{e^{-1} \times 1^1}{1!} = 0.7357$$

(b)

$$P(x>1) = 1 - P(x \le 1)$$

$$= 1 - [P(x=0) + P(x=1)]$$

$$= 1 - 0.7357 = 0.2642$$

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5.11. 20% devices in a laboratory do not work properly. One day 10 devices are selected randomly. Find the probability that, out of the 10 devices, (a) at least 3 work properly, (b) none work properly and (c) 2 to 5 work properly. Find the expected number of devices which work properly.

⇒ In given:—

$p = (100-20)\% = 0.8$

$q = 1 - 0.8 = 0.2$

$n = 10$

(a)

$$P(x \geq 3) = 1 - P(x < 3)$$
$$= 1 - [P(x=0) + P(x=1) + P(x=2)]$$
$$= 1 - [{}^{10}C_0 \times (0.8)^0 (1-0.8)^{10-0} + {}^{10}C_1 (0.8)^1 (1-0.8)^{10-1} + {}^{10}C_2 (0.8)^2 (1-0.8)^{10-2}]$$
$$= 1 - [1.024 \times 10^{-7} + 4.096 \times 10^{-6} + 7.3728 \times 10^{-5})$$
$$= 1 - (7.70 \times 10^{-5})$$
$$= 0.9999$$

(b)

$$P(x=0) = {}^{10}C_0 \times (0.8)^0 (1-0.8)^{10-0}$$
$$= 1.024 \times 10^{-7}$$

(d)

$$P(2 \leq x \leq 5) = P(x=2) + P(x=3) + P(x=4) + P(x=5)$$
$$= 7.3728 \times 10^{-5} + 7.86 \times 10^{-4} + 5.5 \times 10^{-3} + 0.026$$
$$= 0.0327$$

(d)

$$E(x) = np = 10 \times 0.8 = 8$$

5.12. 2.5% signals sent from a server do not reach to its goal properly. A station sends 150 signal per week. Find the probability that, (a) at least 1 do not reach properly, and (b) at best 2 do not reach properly.

⇒ In given:—

$n = 150$

$p = 2.5\%$

$\therefore \lambda = np = 150 \times 2.5\% = 3.75$

(a)

$$P(x \geq 1) = 1 - P(x < 1)$$

$$= 1 - P(x = 0)$$

$$= 1 - \frac{e^{-3.75} \times 3.75^0}{0!}$$

$$= 0.976$$

(b)

$$P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$$

$$= \frac{e^{-3.75}}{0!} + \frac{e^{-3.75} \times 3.75^1}{1!} + \frac{e^{-3.75} \times 3.75^2}{2!}$$

$$= e^{-3.75}\left(1 + 3.75 + \frac{3.75^2}{2!}\right)$$

$$= 0.277$$

Ans:—

5.13. 30% students in AIUB got A^+ grade in Math course last semester. Five students are randomly selected. Find the probability that out of the 5 students, (a) 2 got A^+, (b) all got A^+ and (c) at least 1 got A^+. Find the variance of the no. of students.

⇒ In given:

$$p = 30\% = 0.3$$
$$n = 5$$

(a)

$$P(x=2) = {}^{n}C_{x}\, p^{x}(1-p)^{n-x}$$
$$= {}^{5}C_{2}(0.3)^{2}(1-0.3)^{5-3}$$
$$= 0.441$$

(b)

$$P(x=5) = {}^{5}C_{5}(0.3)^{5}(1-0.3)^{5}$$
$$= 2.43\times10^{-3}$$

(c)

$$P(x\geq1) = 1-P(x<1)$$
$$= 1-P(x=0)$$
$$= 1-{}^{5}C_{0}\times(0.3)^{0}(1-0.3)^{5-0}$$
$$= 0.831$$

Ans:

#Continuous Distribution

Ch: 06

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Continuous Distribution

I = Identification factor

I { average/mean/life span/ quee time/ service time/Length of time

I { mode/wins speed/ power of signal

Exponential P.D

1. $P(X>x)=e^{-x/\lambda}$
2. $P(X<x)= 1-e^{-x/\lambda}$
3. $P(x_1<X<x_2)=e^{-x_1/\lambda}-e^{-x_2/\lambda}$

Rayleigh P.D

1. $P(X>x)=e^{-x^2/2\sigma^2}$
2. $P(X<x)= 1-e^{-x^2/2\sigma^2}$
3. $P(x_1<X<x_2)=e^{-x_1^2/2\sigma^2}-e^{-x_2^2/2\sigma^2}$
4. $\text{mode}=\sigma=\frac{x}{1.253}$

Normal P.D

1. $N(\mu,\sigma^2)$

Example: 1

⇒ Average time needed to open a Computer is 2 min. Find:

(a) Computer will open after 3 min

(b) Computer will open before 2 min

(c) Computer will open in between 2 to 3 mins

(d) Computer will open after the 35 mins (while the man was in a quee for half an hours) (d.2) before 35 mins.

(d.3) within 32 to 35 min.

(a)

$\therefore P(X>3) = e^{-x/\lambda} = e^{-3/2}$
$= 0.2231$

here,
$x = 3$
$\lambda = 2$

(b)

$\therefore P(X<2) = 1 - e^{-x/\lambda}$
$= 1 - e^{-2/2}$
$= 0.6321$

here
$x = 2$
$\lambda = 2$

(c)

$\therefore P(2<x<3) = e^{-x_1/\lambda} - e^{-x_2/\lambda}$
$= e^{-2/2} - e^{-3/2}$
$= 0.1447$

here
$x_1 = 2$
$x_2 = 3$
$\lambda = 2$

(d)

Since, Quee time = 30 min
$\therefore$ he open Computer After $(35-30) = 5$ min

$\therefore P(X>5) = e^{-x/\lambda} = e^{-5/2}$
$= 0.0820$

here
$x = 5$
$\lambda = 2$

(d.2)

$\therefore$ He open computer $(35-30) =$ 5 min before

$\therefore P(X<5) = 1 - e^{-x/\lambda}$

$= 1 - e^{-5/2}$

$= .9179$

here
$x = 5$
$\lambda = 2$

(d.3)

$\therefore$ He will open the computer within $(32-30) = 2$ min to $(35-30) = 5$ min

$P(2<X<5) = e^{-x_1/\lambda} - e^{-x_2/\lambda}$

$= e^{-2/2} - e^{-5/2}$

$= 0.2857$

here,
$x_1 = 2$
$x_2 = 5$
$\lambda = 2$

Ans:-

Examples 6.4:-

$\Rightarrow$ the Average wind speed of a day is 4.5 kmh^{-1} Find the probability that in a randomly selected day, the wind speed

(i) will exceed 4 kmh^{-1}

(ii) will be less than 3 kmh^{-1}

(iii) will be between 2 to 5 kmh^{-1}

$\Rightarrow$ Is given:—

mode $\bar{x} = 4.5$

$$\therefore \sigma = \frac{\bar{x}}{1.253} = \frac{4.5}{1.253} = 3.59$$

(a)

$$P(X>4) = e^{-x^2/\sigma^2 \times 2}$$

$$= e^{-4^2/2\times(3.59)^2}$$

$$= 0.5376$$

here,
$x = 4$
$\sigma = 3.59$

(b)

$$P(X<3) = 1 - e^{-x^2/\sigma^2 \times 2}$$

$$= 1 - e^{-3^2/(3.59)^2 \times 2}$$

$$= 0.2947$$

here,
$x = 3$
$\sigma = 3.59$

(c)

$$P(2<X<5) = e^{-x_1^2/\sigma^2 \times 2} - e^{-x_2^2/2\sigma^2}$$

$$= e^{-2^2/2\times(3.59)^2} - e^{-5^2/2\times(3.59)^2}$$

$$= 0.4772$$

here
$x_1 = 2$
$x_2 = 5$
$\sigma = 3.59$

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Example : 6.5 :—

⇒ The mode of the density of faded out signal is 0.5. Find the probability that the density will be

(i) more than 0.8
(ii) less than 0.4
(iii) between 0.4 to 0.6

(a)

$$P(X>0.8) = e^{-x^2/2\sigma^2}$$
$$= e^{-0.8^2/2\times(0.5)^2}$$
$$= 0.2780$$

here
$\sigma = 0.5$
$x = 0.8$

(b)

$$P(X<0.4) = 1 - e^{-x^2/2\sigma^2}$$
$$= 1 - e^{-0.4^2/2\times(0.5)^2}$$
$$= 0.2738$$

here
$\sigma = 0.5$
$x = 0.4$

(c)

$$P(0.4<X<0.6) = e^{-x_1^2/2\sigma^2} - e^{-x_2^2/2\sigma^2}$$
$$= e^{-0.4^2/2\times(0.5)^2} - e^{-0.6^2/2\times 0.5^2}$$
$$= 0.2039$$

here
$\sigma = 0.5$
$x_1 = 0.4$
$x_2 = 0.6$

Ans:—

Exercise 6:-

6.5. The average time needed to get service in a bank is 15 minutes. Find the probability that a man will be served i) before 25 minutes, ii) between 12 to 18 minutes and iii) after 20 minutes.

⇒ In given:-

$\lambda = 15$

(a)

$$P(X<25) = 1 - e^{-x/\lambda}$$
$$= 1 - e^{-25/15}$$
$$= 0.8111$$

here,
$x = 25$
$\lambda = 15$

(b)

$$P(12<X<18) = e^{-x_1/\lambda} - e^{-x_2/\lambda}$$
$$= e^{-12/15} - e^{-18/15}$$
$$= 0.1481$$

here
$x_1 = 12$
$x_2 = 18$
$\lambda = 15$

(c)

$$P(X>20) = e^{-x/\lambda}$$
$$= e^{-20/15}$$
$$= 0.2636$$

here
$x = 20$
$\lambda = 15$

Ans:-

6.6. The mode of the density of faded out signal is 3.59. Find the probability that the density of a random signal will be i) between 2 to 5 and ii) more than 3.

⇒ In given:-

$\sigma = 3.59$

(I)

$$P(2 < X < 5) = e^{-x_1^2/\sigma^2 \times 2} - e^{-x_2^2/\sigma^2 \times 2}$$

here
$x_1 = 2$
$x_2 = 5$

$$= e^{-2^2/2 \times (3.59)^2} - e^{-5^2/(3.59)^2 \times 2}$$

$$= 0.4771$$

(II)

$$P(X > 3) = e^{-x^2/\sigma^2 \times 2}$$

here
$x = 3$

$$= e^{-3^2/(3.59)^2 \times 2}$$

$$= 0.7052$$

Ans:-

6.7. The average wind speed of a day is 0.6265 (knots). Find the probability that in a randomly selected day, the wind speed will be i) less than 0.5 knots ii) exceed 0.8 knots and iii) between 0.4 to 0.6 knots.

$\Rightarrow$ In given man

$$\bar{x} = 0.6265$$

$$\therefore \sigma = \frac{\bar{x}}{1.253} = \frac{0.6265}{1.253} = 0.5$$

(a)

$$P(X<0.5) = 1-e^{-x^2/2\sigma^2}$$

$$= 1-e^{-(0.5)^2/2\times(0.5)^2} = 0.3934$$

(b)

$$P(X>0.8) = e^{-x^2/2\sigma^2}$$

$$= e^{-(0.8)^2/2\times(0.5)^2} = 0.2780$$

(c)

$$P(0.4<X<0.6) = e^{-x_1^2/2\sigma^2} - e^{-x_2^2/2\sigma^2}$$

$$= e^{-0.4^2/2\times(0.5)^2} - e^{-0.6^2/2\times(0.5)^2}$$

$$= 0.2303$$

TOPIC NAME : TIME : DATE : / /

6.9. The average time needed to send a signal from a server is 5 minutes. Find the probability that a signal will be sent (i) before 310 seconds, and (ii) after 220 seconds.

⇒ In given:—

$$\lambda = 5min = 5 \times 60 = 300 sec$$

(I)

$$P(X < 310) = 1 - e^{-x}$$

$$= 1 - e^{310/300}$$

$$= 0.644$$

(II)

$$P(X < 220) = 1 - e^{-x/\lambda} = 1 - e^{-220/300} = 0.5206$$

6.12. The average time needed to get service in a bank is 15 minutes. If the man waits for 30 minutes in queue, find the probability that he will be served i) within 35 to 50 minutes, ii) after 40 minutes and iii) before 1 hour.

⇒ Is given:—

$\lambda = 15 min$

Queue time = 30 min

GOOD LUCK™

TOPIC NAME : DAY : TIME : DATE : / /

(a)

He get service (35 − 30) = 5 mins to (50 − 30) = 20 mins

$\therefore P(5 < X < 20) = e^{-x_1/\lambda} - e^{-x_2/\lambda}$; $x_1 = 5$, $x_2 = 20$

$= e^{-5/15} - e^{-20/15}$

$= 0.4529$

Ans:

(b)

He get service After (40 − 30) = 10 mins

$\therefore P(X > 10) = e^{-x/\lambda}$; $x = 10$

$= e^{-10/15}$

$= 0.5134$

(c)

He get service before (60 − 30) = 30 mins

$\therefore P(X < 30) = 1 - e^{-x/\lambda}$; here $30 = x$, $15 = \lambda$

$= 1 - e^{-30/15}$

$= 0.8646$

Ans: